Do Recessions Save Lives? Mortality, Persistence, and the Welfare Cost of Business Cycles

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Abstract

We exploit differential geographic intensity of the six major U.S. recessions since 1969 to estimate their effect on mortality and trace implications for the welfare cost of business cycles. We find that a one percentage point increase in a local unemployment rate lowers the age-adjusted mortality rate by 0.66 percent. At the average recession shock of 2.74 percentage points, this implies a 1.8 percent decline in annual age-adjusted mortality. Local projections show the decline persists for more than 25 years. The effect holds across sexes and most age groups, and is concentrated in external causes and mental disorders, with no detectable response for cancer. Age-specific dynamics differ sharply: the mortality response reverts to zero far faster for younger age groups than for older ones. We embed these estimates in a standard model of the lifetime welfare cost of recession risk. Endogenous mortality substantially lowers that cost, allowing the mortality benefit to outlast the recession roughly doubles the offset, and under our calibration recessions become welfare-improving from age 45 onward.

Contents

1 Introduction

Recessions impose costs along many margins. Lost output, unemployment spells, foregone earnings, and scarred careers are all well measured, and they discipline how macroeconomists calculate the welfare cost of business cycles [Lucas, 1987, İmrohoroğlu, 1989, Storesletten et al., 2001, Krebs, 2007]. Yet there is evidence that mortality falls when the economy contracts [Ruhm, 2000, Stevens et al., 2015, Finkelstein et al., 2025], suggesting that the welfare costs of recessions emphasized in the macroeconomics literature may be too narrowly focused; incorporating the mortality impacts of recessions has important implications for their welfare consequences, both in the aggregate and across demographic groups.

This paper studies the mortality effects of the six major U.S. recessions since 1969, drawing on data through 2016. First, we independently replicate the Great Recession analysis of Finkelstein et al. [2025]. We then extend the strategy—leveraging spatial variation in the economic severity of recessions—to the full set of six, providing new empirical evidence on the contemporaneous and long-run impact of recessions on mortality, and on whether that impact has remained stable across decades.

Across our analyses, we corroborate the finding that the Great Recession greatly reduced mortality and establish a robust pattern of this behavior over time. For every one-percentage-point increase in a Commuting Zone's (CZ) unemployment rate between 2007–2009, its age-adjusted mortality rate fell by 0.59 percent, with an immediate response and persistence through ten years. On average across the six recessions, we find that equivalently constructed unemployment shocks at the state level correspond with a 0.66 percent drop. Because recessions occur close together in our historical panel, we cannot observe as many post-recession years for the other five events as the Great Recession replication provides, but we confirm the prompt drop in mortality as robust across all six. Since the average national unemployment rate increased by 2.74 percentage points across recessions, our estimates imply that the magnitude of the reduction in annual age-adjusted mortality is 1.8 percent.

Our analysis also extends the observable horizon well beyond the ten-year window of the existing design, using local projections to leverage the length of our panel [Jordà, 2005]. We find the response of age-adjusted mortality remains negative for over 25 years, with an average 0.54 percent drop per percentage point of shock. We estimate a variety of other models that provide support for the length and magnitude of this shockingly persistent effect across business cycles.

We look into heterogeneity along a variety of dimensions—cause of death, age, race, and sex—and find broadly consistent patterns across our longer panel. Mortality declines are largest and most persistent among older age groups, while the youngest (under 25) and causes such as mental disorders are highly cyclical. Cardiovascular mortality shows a persistent but noisily estimated decline, whereas external causes respond more contemporaneously. Our extended horizon further reveals sharp differences in age-specific dynamics: the mortality response reverts to zero substantially faster for younger age groups than for older ones, consistent with acute external causes tracking economic conditions while chronic causes generate durable health effects.

Identifying the channels through which recessions reduce mortality is complicated by the fact that they shift across decades—the composition of the labor force has changed dramatically [Goldin and Katz, 2002, Feyrer, 2007, Mennuni, 2019], the structure of the healthcare system has transformed [Cutler and McClellan, 2001, Cutler, 2004, Gaynor et al., 2015], and the leading causes of death have evolved [Becker et al., 1994, Cawley, 2015]. We leave a full decomposition to future work and instead focus on folding our generalized estimates into a welfare framework.

Our welfare analysis follows the framework of Finkelstein et al. [2025], performing two exercises to quantify the welfare effects of recessions when factoring in mortality declines. In the first, we extend the Krebs [2007] model of the consumption-based welfare cost of facing a lifetime risk of recessions by introducing mortality—both exogenous background mortality and endogenous pro-cyclical mortality. Our differentiation is to allow the endogenous component to persist beyond recession periods, consistent with our empirical findings.

In the second exercise, we isolate the impact of a single recession, calibrating a ten-year downturn—consistent with our cross-sectional estimates—with a mortality effect that persists through year 25. This allows us to quantify how the welfare offset from reduced mortality grows when persistence is accounted for. In each, we find that the inclusion of this sticky endogenous mortality substantially lowers the welfare costs of recessions.

Our results have various limitations. First, because we control for area, time, and recession fixed effects, we estimate how much more mortality fell in harder-hit areas, not what the recession did to mortality nationally. Our welfare exercise nonetheless applies this differential as an economy-wide discount, following Finkelstein et al. [2025]. Second, while we capture a variety of recessions, they are all national events, so our results may not generalize to localized crises. However, the consistency of our results across unique recessions suggests a higher level of generalizability than we could find from looking at any single recession. Third, and in the same vein, our estimates, especially in our long term panel, may miss local health effects due to only being at the state level, and are measured on populations that migrate in response to the shock itself [Arthi et al., 2022], as well as all of our analyses featuring mortality, rather than any less severe health outcomes, as the sole estimate. Finally, although we leverage our temporal window to provide estimates that extend further than the scope of a single business cycle, we cannot guarantee that these are free of any bias due to correlation across recessionary events. Recognizing these constraints, our paper offers new evidence for the existence, features, and dynamics of recession-induced mortality declines and posits that accounting for this nontraditional recession impact notably alters the welfare costs associated with economic downturns.

Our paper furthers the novel work of Finkelstein et al. [2025], who give unique attention to quantifying the welfare effects of cyclical volatility in health, in investigating the relationship between mortality and recessions [Ruhm, 2000, 2003, Miller et al., 2009, Ruhm, 2015, Stevens et al., 2015]. That literature is largely built on contemporaneous fixed-effects regressions, which leaves open both whether the relationship is stable across cycles and whether it reflects a durable improvement in population health or the harvesting of the most medically fragile. Evidence across scopes is mixed: Coile et al. [2014] find short-run mortality reductions for near-retirement workers but longevity losses from sustained employment loss, and Sullivan and von Wachter [2009] show that displacement persistently elevates mortality among prime-age men. Additionally, local exposure to manufacturing decline and import competition raises mortality [Autor et al., 2013, Pierce and Schott, 2020], while a growing share of the divergence in mortality across areas loads on behavioral factors—smoking, obesity, and substance abuse—rather than income directly [Becker et al., 1994, Cawley, 2015, Case and Deaton, 2015, Currie and Schwandt, 2016].

On the aggregate side, we contribute to a large empirical literature on the relationship between health and the economy. Among developed countries the negative relationship between income and mortality is consensus, in the cross section and within countries over time [Preston, 1975, Deaton, 2003, Cutler et al., 2006, Chetty et al., 2016]. Outside of the United States, comparable pro-cyclical patterns appear in Neumayer [2004] for Germany and Ariizumi and Schirle [2012] for Canada, though the latter finds no response for infants or the elderly.

Taken together, this evidence—secular and economy-driven heterogeneity, aggregate versus individual effects—leaves it unclear ex ante what to expect from differential exposure to recessionary shocks. Empirically, we use a single shock to an area's unemployment rate—in the tradition of Bartik [1991] and Blanchard and Katz [1992], and as in Yagan [2019]'s treatment of Great Recession employment hysteresis—rather than an area's contemporaneous unemployment rate, allowing the mortality response to be lagged [Finkelstein et al., 2025]. We then show that this identification extends across a pool of recessions, and we combine the same shock with local projections [Jordà, 2005] to trace persistence beyond the horizon a single cycle can support. Finally applying our empirical results to a welfare model, our work fits into the macroeconomics literature on the welfare cost of business cycles [Lucas, 1987, İmrohoroğlu, 1989, Storesletten et al., 2001, Barlevy, 2004, Krebs, 2007, Krusell et al., 2009].

The rest of the paper is organized as follows. Section 2 lays out the data and empirical strategy. Section 3 presents the main results. Section 4 takes the empirical results to welfare calculations. Section 5 concludes.

2 Data and Empirical Strategy

2.1 Data

For our analysis, we concentrate on the United States, the fifty states plus the District of Columbia, and the years 1969 to 2016. In our first exercise, we independently replicate the analysis of Finkelstein et al. [2025], focusing on the Great Recession and its aftermath from the years 2003-2016, beginning the analysis in 2003 to avoid confounding effects from the 2001-2002 recession [Yagan, 2019]. We do this primarily for Commuting Zones (CZs), which are standard aggregations of counties designed to capture local labor markets, but we also establish consistency of our results at the state and county levels. This gives us precedent to extend the temporal scope of our analysis for the following exercises, covering the years 1969-2016, spanning an additional five major recessions. For this, we are limited to analysis at the state level due to the lack of availability of county-level economic indicators before the 1990s. We provide an overview of our main data sources here; Appendix A describes our data construction and variable definitions in more detail.

Mortality. Our source of mortality is in line with the previous literature, as we use publicly available data from the Centers for Disease Control and Prevention's (CDC) WONDER database. For robustness, we use both the CDC-given age-adjusted mortality rate (AAMR) for a given geography, as well as creating our own AAMR values by combining raw deaths data from WONDER with population data from the National Cancer Institute's (NCI) Surveillance, Epidemiology, and End Results (SEER) Program and 2000 US Standard Age Weights from the CDC. For each geography, we construct also AAMR by race, sex, and cause of death, as well as age-specific mortality. Figure 1 shows the dynamics of national AAMR over our panel.

Economic indicators. We use a variety of economic indicators across our analyses, under the constraint of data availability. In our first Great Recession analysis, we contruct geography-year unemployment rates and employment to population (EPOP) ratios using the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS) and geography-year real gross domestic product (GDP) per capita using data from the Bureau of Economic Analysis (BEA) and SEER. For our longer term analysis, we use two sources for unemployment. For our earliest years, 1969-1976, covering roughly the first two recessions in our period, we use claims-based predicted state unemployment rates from Fieldhouse et al. [2024] followed by BLS state-level aggregate unemployment rates for the remainder, 1977-2016. Similarly for real GDP per capita, we rely on the previous literature, combining state-level GDP data from Kleinman et al. [2023] with SEER population data; we are able to use this for all years 1969-2016.

Other data sources. We supplement our main analysis with data from the IPUMS Current Population Survey Annual Social and Economic Supplement (CPS ASEC) to construct state-level measures of household income, although some states are aggregated for consistency with early reporting.

Table 1 presents summary statistics for our main variables across both the Great Recession and full panel samples.

2.2 Measuring Recession Shocks

Following the literature, we parameterize the severity of the Great Recession in area c, $SHOCK_c$, as the percentage point change in the area unemployment rate between 2007 and 2009—the peak-to-trough change over the recession [Blanchard and Katz, 1992, Autor et al., 2013, Yagan, 2019, Finkelstein et al., 2025]. For each of the other five recessions in our historical panel, we parameterize $SHOCK_i^r$ analogously, as the state-specific peak-to-trough change in the unemployment rate around recession r.

Figure 13 plots the distribution of $SHOCK_c$ across 741 commuting zones for the Great Recession. The mean shock is 3.9 percentage points, with substantial spatial variation: areas in the top quartile experienced average increases of 6.6 percentage points compared to 1.5 in the bottom quartile. This variation extends to the full panel, as Nevada, Florida, and California experienced shocks exceeding 6 percentage points, while North Dakota, Nebraska, and South Dakota saw increases below 2 percentage points. Because our panel spans six recessions over nearly five decades, we can also examine how shock magnitudes have moved over time. Figure 15 plots the mean state-level shock for each recession cohort: the Great Recession (4.1 pp) and the 1980–82 double-dip (3.7 pp)[1] produced the largest shocks, while the 1990–91 and 2001–02 recessions were considerably milder (1.7 and 1.5 pp, respectively). This cross-cohort variation in shock magnitudes ensures our estimates are not driven by any single episode.

Our use of the unemployment rate to parameterize recession severity follows the existing literature on recessions and mortality [Ruhm, 2000, 2003, Stevens et al., 2015]. To confirm

[1] Because the 1980 and 1981–82 recessions are separated by only one year, we combine them into a single cohort throughout; this is why our panel spans six recession cohorts rather than seven.

that the shock captures broad-based economic distress rather than labor market churn alone, we examine the response of other economic outcomes: the employment-to-population (EPOP) ratio and real GDP per capita in the Great Recession, and real GDP per capita and household income in the full historical panel. Each falls in areas hit by larger unemployment shocks (Figure 16).

2.3 The Great Recession

We follow Finkelstein et al. [2025] and Yagan [2019] in exploiting spatial variation in the severity of the Great Recession. Our main estimating equation is:

$$\log \mathrm{AAMR}_{ct} = \beta_t \big[SHOCK_c \times \mathbf{1}(Year_t)\big] + \alpha_c + \gamma_t + \varepsilon_{ct} \tag{1}$$

where c indexes an area, $\log \mathrm{AAMR}_{ct}$ is log age-adjusted mortality in area c and year t, $SHOCK_c$ is a measure of the local severity of the Great Recession, $\mathbf{1}(Year_t)$ is an indicator for calendar year t, and α_c and γ_t are area and year fixed effects. Our primary unit of analysis is the commuting zone; we also estimate equation (1) at the county and state level to establish robustness across geographic units. We estimate equation (1) by OLS and cluster standard errors at the level of c. The coefficients of interest are the $\{\beta_t\}$, which trace the dynamic response of log AAMR to a one percentage point increase in local recession severity in each calendar year t. We omit the interaction with $SHOCK_c$ in 2006, so all β_t are relative to that pre-recession baseline; our sample runs from 2003 to 2016, an event-time window of $t - 2006 \in \{-3, \dots, +10\}$.

Our key identifying assumption is that no shocks to mortality coincide with the timing of the Great Recession and are correlated with the local severity of its economic impact. The pre-recession coefficients provide a direct test of this assumption.

2.4 The Full Recession Panel

The Great Recession event study covers a single episode—one driven by an unprecedented housing and credit crisis [Brunnermeier, 2009, Reinhart and Rogoff, 2009] with geographically concentrated employment losses [Mian and Sufi, 2014]. To assess whether the mortality response is a stable feature of modern U.S. business cycles rather than an artifact of these unique conditions, we again exploit spatial variation in local recession severity, now pooling across all six major U.S. recessions since 1969 in a stacked event study; data availability restricts this analysis to the state level. We construct a dataset in which each observation is identified by state i, recession r, and event time $\tau = t - T_0^r$, where T_0^r is the year before recession r's onset, and estimate:

$$\log \text{AAMR}_{i\tau r} = \beta_\tau \big[SHOCK_i^r \times \mathbf{1}(EventTime_\tau)\big] + \alpha_{ir} + \lambda_{\tau r} + \varepsilon_{i\tau r} \tag{2}$$

where $SHOCK_i^r$ is a measure of the local severity of recession r, α_{ir} are state-by-recession fixed effects, and $\lambda_{\tau r}$ are event-time-by-recession fixed effects. We omit the interaction with $SHOCK_i^r$ at $\tau = 0$, so all β_τ are relative to each cohort's pre-recession baseline, and cluster standard errors at the state level. The pooled coefficients $\{\beta_\tau\}$ capture the average dynamic response across recessions. The identifying assumption is the same as above, applied within each cohort: no shocks to mortality coincide with a given recession's timing and are correlated with its local severity.

Window Specifications. We estimate two variants of equation (2). The first uses a constant window of $\tau \in \{-3, \ldots, +5\}$ for all recessions, keeping pre- and post-periods directly comparable across cohorts. The second uses recession-specific windows, with window length varying by recession to prevent overlap between adjacent events. Our preferred specification uses the non-overlapping windows, as they maximize the observable post-recession horizon for each cohort without contaminating the estimates with shocks from adjacent recessions. We present the constant-window results as a robustness check to verify that the pooled estimates are not driven by window construction. Table 8 in the Appendix reports, for each recession cohort, the calendar years spanned by each specification.

2.5 Local Projections

To assess whether the mortality effect persists beyond the event study window—whose length is set by the spacing of recessions rather than by the data—we estimate local projections [Jordà, 2005] pooling all six recession cohorts:

$$\log \text{AAMR}_{i,T_0^r+h} = \alpha_i^h + \gamma_r^h + \beta^h \cdot SHOCK_i^r + \sum_{l=1}^{3} \phi_l^h \cdot \log \text{AAMR}_{i,T_0^r-l} + \sum_{l=1}^{3} \psi_l^h \cdot UR_{i,T_0^r-l} + \varepsilon_{ir}^h \tag{3}$$

where T_0^r is recession r's reference year, $h = 0, 1, \ldots, 25$ indexes years since that reference year, α_i^h and γ_r^h are state and recession fixed effects, and three lags each of log AAMR and the unemployment rate control for pre-existing dynamics. We estimate a separate regression at each horizon. Standard errors are clustered at the state level.

2.6 Mortality Patterns Across Geography and Time

Before presenting results, we examine whether raw mortality patterns are consistent with our identifying assumption. For the Great Recession, we compare the spatial distribution of $SHOCK_c$ to each area's pre-recession mortality rate (Figure 17), and we plot age-adjusted mortality separately for areas in the top and bottom quartiles of $SHOCK_c$ from the years preceding the recession through the end of our sample (Figure 18). High- and low-shock areas show no pre-recession divergence.

Our historical panel allows a more demanding version of this check. Because we observe six recession cohorts spanning nearly five decades, we compare high- and low-shock states across the full 1969–2016 window rather than around a single recession (Figures 19 and 20). High- and low-shock states track each other closely across the full window, and within each cohort a state's pre-recession mortality is largely uncorrelated with the shock it goes on to experience.

3 Mortality Impacts of Recessions

Here, we exhibit estimates of the recessionary effects on mortality, both overall and across various subgroups and causes of death, for both the Great Recession and the six recession panel. Outside of the main results from the Great Recession, we will focus most heavily on our stacked event study and local projection results; we include full heterogeneity analysis for the Great Recession in Appendix C.

3.1 Overall Mortality Estimates

Baseline estimates. We begin by replicating the results of Finkelstein et al. [2025]. Figure 3 presents estimates of equation (1) at the commuting zone and state levels, with the coefficient on β_{2006} normalized to zero [2]. Similar to them, we find an immediate and persistent—at least ten-year—decline in log age-adjusted mortality. The timing is consistent with local economic indicators also beginning to deteriorate in 2007. We also detect a slight positive pre-trend in mortality before 2006, opposite in sign to the post-recession decline, alongside a relative improvement in economic indicators in hard-hit areas over the same pre-period; this suggests our estimates may understate the true extent of the recession-induced mortality decline.

Extending our scope to include the previous five recessions, we find results consistent with our Great Recession-specific findings using equation (2), shown in Figure 4. Although the time between recessions varies across our panel, and thus we do not observe a full ten years

[2] County-level estimate is found in Appendix Figure 23

of effects for any other recession, we find the same qualitative pattern: an immediate and persistent decline in mortality beginning at recession onset. The pattern holds under both window specifications.

For the Great Recession, log AAMR falls sharply and remains negative through 2016 — ten years after the baseline. The average coefficient over 2007–2009 is −0.59, and over the full post-recession window (2007–2016) is −0.72. With an average state-level unemployment shock of 4.6 percentage points, these estimates imply a roughly 3.3 percent decline in age-adjusted mortality attributable to the Great Recession. These results are qualitatively consistent with Finkelstein et al. [2025], who find an average coefficient of approximately −0.5, implying a 2.3 percent decline. We find somewhat larger magnitudes, with the deviation concentrated at larger geographies (state and CZ rather than county), likely due to our use of publicly available aggregate mortality data rather than the restricted microdata; we test whether mortality reductions persist in areas whose labor markets have recovered by 2016 by splitting CZs at the median of EPOP recovery—defined as the fraction of the 2006–2009 EPOP decline regained by 2016—and re-estimating the event study separately for each group. CZs with above-median recovery show average coefficients of −0.44 over 2007–2009 and −0.52 over 2007–2016; those with below-median recovery show −0.51 and −0.35, respectively. Both groups exhibit persistent mortality declines of similar magnitude, indicating that the effect is not merely a contemporaneous correlate of current labor market slack. Figure 21 confirms that the two groups do in fact diverge in EPOP trajectories: above-median recovery CZs see EPOP rebound toward pre-recession levels, while below-median CZs do not, yet both exhibit persistent mortality declines; this check is GR-specific due to data availability.

Figure 4 is the main stacked event study, pooling all six major U.S. recessions since 1969. The pre-recession coefficients drift mildly, consistent with the secular decline in age-adjusted mortality; we do not read the pre-period as flat. Immediately following recession onset, log AAMR falls sharply. The average post-recession coefficient over horizons 1–10 is −0.66, statistically significant at nine of the ten post-recession horizons. Pooling six cohorts of state-level variation buys breadth at some cost in precision, and we do not claim the pooled estimate is robust to any violations of parallel trends. What the pooled specification establishes is the shape and persistence of the response, as it closely mirrors the Great Recession-specific estimates, though the pooled magnitude is smaller than the state-level GR average of −1.02. The mortality response is not specific to the 2007–2009 episode but appears to be a recurring feature of U.S. business cycles over the past five decades [3].

[3] We include robustness checks in the form alternative specifications (Inverse Hyperbolic Sine and Levels) and of a constant window event study and "leave-one-out" for the Great Recession in the Appendix Figure 24, Figure 38.

Combining the pooled coefficient of -0.66 with the average shock of 2.74 percentage points across our panel, a typical recession's peak-to-trough unemployment increase implies a roughly 1.8 percent decline in age-adjusted mortality.

3.2 Long-Run Effects: Local Projections

The event study horizon is bounded by the spacing of recessions, and the effect does not turn around within it: the tenth horizon coefficient in Figure 4 is -0.79, larger in magnitude than the -0.66 average. But how long the decline lasts matters both for the welfare calculation in Section 4, which scales directly with duration, and for whether the decline reflects fewer deaths or merely later ones. Local projections allow us to extend the horizon to twenty-five years.

Figure 5 plots the local projection impulse responses of both the unemployment rate and log AAMR over a 25-year horizon. The unemployment rate response rises on impact and returns to zero within ten years, confirming that the labor market shock itself is transitory. The AAMR response is not: it is persistently negative, stabilizing at around -0.5 log points per unit shock—a 1.5 percent decline in mortality at the average recessionary shock of 2.9 percentage points—and only gradually returns toward zero late in the horizon. The two clocks decouple—mortality remains depressed for roughly fifteen years after unemployment has recovered—a dynamic not seen in the event studies. Confidence intervals are wide throughout, so we do not emphasize the precise magnitude at any single horizon; what the exercise establishes is the sign and the duration. The response also does not reverse at any horizon, which provides evidence against pure mortality displacement.

Reassuringly, the first ten horizons track the event study estimates closely. This is worth more than a consistency check, because the two specifications do not rest on the same identifying assumption: the event study requires parallel trends, while equation (3) conditions on three lags of log AAMR and the unemployment rate, so β^h is identified off shock exposure given each state's pre-recession level and trajectory. Two designs with different assumptions delivering the same short-run response is a stronger argument than either alone, and it is an appropriate response to the mild pre-trend discussed in Section 3.1.

We verify that this lag structure is sufficient by re-estimating with 5, 7, and 10 lags [4] (Appendix Figure 22); results are qualitatively and quantitatively similar across all specifications, indicating that the impulse response is not sensitive to the degree of dynamic control.

Our LP provides aggregate-level evidence of a long-run mortality benefit operating through a distinct channel—the broad reduction in economic activity and its associated externalities—

[4] We also include the LP with GDP per capita shock, showing that the qualitative response is consistent with other economic indicators.

that persists well beyond the recession window. Taken together, the event study and local projection results are consistent with recessions having genuine persistent effects on population health, not merely contemporaneous impact or a transient rearrangement of the timing of deaths.

The persistence itself narrows the candidate mechanisms, even though we do not attempt to formally identify one. A channel operating through current economic conditions cannot generate a mortality response that outlasts the unemployment response by fifteen years, and the EPOP-recovery split of Section 3.1 points the same way. What survives is something that changes a stock rather than a flow—cumulative exposure, or a persistent shift in behavior or environmental conditions—leaving a lasting imprint from a temporary contraction.

3.3 Unpacking the Overall Mortality Decline

We now decompose the overall mortality decline along three dimensions—cause of death, age, and demographics—to understand which subgroups drive the aggregate result. Figure 7 provides a summary of average coefficients across each overall mortality source for the Great Recession and six-recession panel to establish robustness.

By Cause of Death. Figure 8 presents average coefficients by cause of death (full event studies Figure 26, Figure 27). External causes show the largest and most consistently significant response: an average coefficient of −0.80 in 2007–2009 growing to −2.49 over 2007–2016 in the GR, and −1.50 in the pooled panel. Mental disorders exhibit similarly large point estimates (−1.58 in the GR short run, −4.71 in the long run, and −2.68 pooled), though with wider confidence intervals. Cardiovascular and metabolic mortality declines significantly in the GR short run (−0.89) but attenuates over the longer horizon. Cancer—the second leading cause of death—shows no significant response in either sample, consistent with literature on the subject. Digestive and genitourinary causes decline significantly in the pooled panel (−0.93). Neurological mortality shows no significant effect or, if anything, a slight increase.

By Age. Figure 9 presents average coefficients by age group (full event studies Figure 28, Figure 29, Figure 30, Figure 31). In the Great Recession, the mortality decline is broad-based across age groups, with particularly large effects among children aged 1–4 (−4.12 in 2007–2009) and the elderly aged 75–84 (−0.88). The six-recession panel reveals a similar pattern: the decline is significant across most age groups, with the largest effects among infants (−1.41), young children (−2.34), and young adults aged 25–34 (−1.87). The 55–64 age group is a notable exception, showing small and insignificant effects in both samples, possibly reflecting offsetting forces, possibly indicating reduced occupational hazard but also reduced access to employer-sponsored health insurance upon job loss.

By Race and Sex. Figures 10 and 11 examine heterogeneity by race and sex (full event studies Figure 32, Figure 33, Figure 34, Figure 35). By race, the effect is concentrated among the White population, whose event study closely mirrors the pooled result. The Black population shows no statistically significant treatment effect at any post-recession horizon. This racial heterogeneity is notable and could reflect differences in labor market exposure, occupational composition, residential proximity to pollution sources, or access to healthcare, though identifying the precise channel is beyond the scope of this paper. By sex, the results are broadly consistent across males and females. Both groups exhibit flat pre-trends and statistically significant post-recession declines, with coefficients ranging from approximately -0.2 to -0.4 across post-recession years. The similarity in magnitude across sexes suggests the effect is not driven primarily by industries with heavily male workforces.

3.4 Sensitivity Analysis

Population Flows. A concern with area-level mortality studies is that recessions may induce selective migration, confounding the estimates if healthier or sicker individuals differentially leave harder-hit areas [Arthi et al., 2022]. Finkelstein et al. [2025] address this using individual-level panel data to track movers; we lack such data but conduct an aggregate check by estimating our event studies with log population as the outcome (Appendix Figure 36). Consistent with their results, we find slightly upward-sloping pre-trends—harder-hit areas were gaining population faster before the Great Recession—but flat post-recession coefficients, indicating no differential population response to the shock. Our full panel results see a more pronounced dip, but it is most insignificant. This suggests that selective migration is not driving our mortality results.

Geographic Unit. For the Great Recession analysis, our results are qualitatively consistent across the county, CZ, and state levels, though coefficients drift more negative at larger geographies—likely reflecting our use of publicly available aggregate data rather than the restricted microdata employed by Finkelstein et al. [2025]. The qualitative message remains the same or, if anything, strengthens at broader geographic scales. For the historical panel, data availability restricts us to the state level.

Linearity in the Shock. The pooled specification imposes a linear dose-response, and two checks suggest the magnitude is sensitive to it. Adding quadratic shock interactions rejects the null that they are jointly zero, and trimming the decile of state-recession cells with the largest shocks attenuates the pooled coefficient from -0.66 to -0.48. Both bear on the size of the response rather than its sign. To separate the two, we replace the continuous shock with an indicator for an above-median within-recession shock, which compares groups of states without assuming linearity. States with above-median shocks see log AAMR fall by 0.012 relative to below-median states over the ten post-recession years, significant at six of ten horizons. Estimating the same contrast on the trimmed sample—relaxing linearity and dropping the largest shocks at once—leaves the response negative at every post-recession horizon (Figure 37). We therefore read the direction and persistence of the pooled result as robust to functional form, and treat the coefficient itself as a summary of recessions of the magnitudes we observe rather than a per-unit effect to be rigorously extrapolated.

4 Welfare Model

In order to quantify our results, we incorporate these mortality declines into welfare calculations. This is in line with the literature, assuming that mortality impacts resulting in longer life expectancy translate to improvements in well-being [Murphy and Topel, 2006, Becker et al., 2005, Jones and Klenow, 2016]. In principle we replicate the two exercises in Finkelstein et al. [2025]: first, where they incorporate endogenous mortality into Krebs [2007]'s model of the welfare cost of business cycles, and second, where they implement a one-time, ten-year recession into an otherwise recession-free economy. But in each case we add in a persistence in the mortality benefit that lasts past the economic recovery, in line with our empirical results. The two exercises share a utility function, an income process, a welfare criterion, and a calibration; they differ only in the path of the aggregate state and in how mortality is indexed to it. We set out the common parts first, then take each exercise in turn.

4.1 Setup

Utility. We consider a representative agent whose expected lifetime utility is given by:

$$U(c(t), m(t)) = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t S(m(t)) u(c(t)) \right] \tag{4}$$

where $c(t)$ is consumption in period t, $m(t)$ is the (age-specific) mortality rate, and β is the discount factor. The cumulative survival probability is $S(m(t)) = \prod_{\tau=0}^{t}(1-m(\tau))$. Per-period utility follows the constant-relative-risk-aversion form used by Hall and Jones [2007]:

$$u(c) = b + \frac{c^{1-\gamma}}{1-\gamma} \tag{5}$$

where γ is the coefficient of relative risk aversion and b governs the value placed on additional years of life. We report results across a range of γ; welfare costs are increasing in γ. Since $u'(c) = c^{-\gamma}$, the value of a statistical life-year (VSLY) implied by this utility function is $\text{VSLY} = u(c)/u'(c) = bc^{\gamma} + c/(1-\gamma)$. We calibrate b to match a target VSLY at reference consumption c: $b = \text{VSLY} \cdot c^{-\gamma} - c^{1-\gamma}/(1-\gamma)$.

Recessions and Income Processes. The aggregate state $\omega \in \{L, H\}$ follows a stochastic process, and income evolves according to Krebs [2007]: income is subject to persistent idiosyncratic shocks and to job displacement risk, with both the probability of displacement and the earnings loss conditional on displacement higher in the bad aggregate state ($\omega = L$) than in the normal state ($\omega = H$). Since the agent is assumed to consume all income each period, $c(t) = y(t)$, consumption responds to the aggregate state purely contemporaneously. We make no change to this part of the model. Appendix B.1 reproduces the full income process.

The Welfare Cost of Recessions. We define the welfare cost of recessions, Δ^{dm}, as the amount the agent would need to be paid, as a percentage of average annual consumption, to accept an economy in which recessions occur relative to an otherwise identical economy that remains in the no-recession state $\omega = H$ for all t:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t S(m(t)) u\big((1+\Delta^{dm})c(t)\big) \right] = \mathbb{E}_0^{\omega=H} \left[\sum_{t=0}^{\infty} \beta^t S(m^H(t)) u(c(t)) \right] \tag{6}$$

The left-hand side is expected lifetime utility in the recession economy, where $m(t)$ is the age-specific mortality path implied by that exercise's mortality process; the right-hand side is expected lifetime utility in an economy that remains in the no-recession state throughout, where $m^H(t)$ is the age-specific mortality rate in normal times, taken from the 2007 Social Security Administration period life table [Social Security Administration, 2010]. Expectations are taken over the idiosyncratic income and displacement risk of the Krebs [2007] process, which is common to both sides. Everything that follows changes only $m(t)$.

Mortality. Finkelstein et al. [2025] incorporate endogenous mortality by assuming a recession lowers the mortality rate by a constant percentage across all age groups only while the aggregate state is currently bad: $m^L(t) = (1+dm) \cdot m^H(t)$ if $\omega_t = L$, and $m^H(t)$ otherwise, with $dm < 0$. This assumes the mortality benefit of a recession vanishes the instant the aggregate state reverts to normal. Because our local projection estimates show the mortality response to a recession persists well after the shock itself has dissipated (Section 3.2), this contemporaneous assumption may understate the true welfare offset from recession-induced mortality declines. In each exercise below we take their specification as the benchmark and depart from it only in how long the discount dm applies.

We calibrate dm in two ways. First, we hold dm at the value used by Finkelstein et al. [2025], so that the only difference between our estimates and theirs is the duration of the discount. This isolates the all-else-equal effect of letting the mortality benefit outlast the recession. Their calibration combines the Great Recession unemployment shock of 4.6 percentage points with a mortality response of -0.5 log points per percentage point, giving $dm = -0.023$. Second, we use our own pooled estimate from the stacked event study, which reflects a typical recession over the past five decades rather than the Great Recession alone: an average shock of 3.2 percentage points and a response of -0.69 log points per percentage point sustained over a twenty-year horizon, twice the length of the recession itself, giving $dm = -0.022$. The two values are close, so the choice of calibration has little bearing on the results; the persistence, not the magnitude of dm, drives the difference. We use $dm = -0.023$ throughout.

Calibration. We adopt the calibration of Finkelstein et al. [2025] throughout. We report results for risk aversion $\gamma \in \{1.5, 2, 2.5\}$, for VSLY values of \$100,000, \$250,000, and \$400,000, and for agents starting at ages 35, 45, 55, and 65, extending to 75 in the Great Recession exercise, where the mortality channel operates most strongly. We take the income process parameters from Krebs [2007] unchanged, including the annual discount factor $\beta = 0.99$.

4.2 The Welfare Cost of Recessions

4.2.1 Sticky Mortality

The aggregate state ω_t evolves stochastically, as in Krebs [2007]. As a first, minimal departure from the contemporaneous benchmark, we let the mortality discount persist for one additional period after a recession ends, without allowing successive recessions to compound the discount. Formally, define

$$\tilde{L}_t = \mathbf{1}(\omega_t = L) \vee \mathbf{1}(\omega_{t-1} = L) \tag{7}$$

and let

$$m(t) = \begin{cases} (1 + dm) \cdot m^H(t) & \text{if } \tilde{L}_t = 1 \\ m^H(t) & \text{if } \tilde{L}_t = 0 \end{cases} \tag{8}$$

so that a single recession year now generates two years of reduced mortality, but consecutive or repeated recessions do not compound the discount beyond $(1 + dm)$. Setting $\tilde{L}_t = \mathbf{1}(\omega_t = L)$ recovers the contemporaneous specification exactly as a special case.

4.2.2 Results

Table 4 reports the welfare cost of business cycles, Δ^{dm}, as a percentage of average annual consumption. Column (1) holds mortality exogenous. The cost is positive at every age and rises with risk aversion, from 3.06 percent at age 35 under $\gamma = 1.5$ to 4.49 percent under $\gamma = 2.5$. It also falls with age, since older agents face fewer remaining years of labor market risk: at $\gamma = 2$ the cost drops from 3.73 percent at age 35 to 0.86 percent at age 65.

Columns (2)–(4) apply the contemporaneous specification of Finkelstein et al. [2025] and reproduce their central finding: endogenous mortality substantially lowers the welfare cost of business cycles, and the offset grows with both the VSLY and the agent's age. At age 65 recession risk is welfare-improving under every VSLY we consider.

Columns (5)–(7) hold dm fixed and add only the persistence of equation (8), so a single additional year of reduced mortality per recession year is the only difference between the two sets of columns. That one year roughly doubles the mortality offset. At age 35 and $\gamma = 1.5$, a $400,000 VSLY lowers the cost from 3.06 to 1.83 percent contemporaneously, and to 1.23 percent once the discount persists; at age 65 and $\gamma = 2.5$ the same VSLY moves the cost from -3.82 to -6.03 percent. Persistence also lowers the age at which recessions become welfare-improving, from 55 to 45. Figure 12 traces all three specifications across starting ages from 35 to 75, holding $\gamma = 2$ and the VSLY at $250,000. The sticky path lies below the contemporaneous path at every age, and the vertical gap between them widens as the agent ages.

The pattern in γ reverses with age in both specifications: at younger ages higher risk aversion raises the net cost, since consumption risk dominates, while at 65 it deepens the gain, since the mortality benefit dominates and is itself valued more highly. Persistence widens this reversal, emphasizing a unique distributional implicaition of welfare, as young people still dislike recessions, but older people may strongly prefer to be in a bust.

4.3 Welfare Analysis of the Great Recession

4.3.1 Decoupled Clocks

What changes in this exercise is the aggregate state. Rather than letting ω_t evolve stochastically, we impose a single recession. A labor market shock hits at $t = 0$ and lasts for $N_c = 10$ years, after which the economy permanently reverts to its pre-recession state and never leaves it:

$$\omega_t = L \text{ for } 0 \leq t \leq N_c - 1, \qquad \omega_t = H \text{ for } t \geq N_c \tag{9}$$

Displacement probabilities and earnings losses take their bad-state values over the first N_c years and their normal-state values thereafter. We make no change to N_c or to the path of ω_t.

Our departure is in how long mortality responds to this shock. They tie the mortality decline to the economic state directly, applying the constant percentage decline dm for exactly as long as the economy remains in the recession state $\omega_t = L$, i.e., for $N_c = 10$ years. We keep the decline constant but extend it over the horizon implied by our local projections, decoupled from the duration of the labor market shock:

$$m(t) = m^H(t) \cdot (1 + dm) \text{ for } 0 \leq t \leq N_m - 1, \qquad m(t) = m^H(t) \text{ for } t \geq N_m \tag{10}$$

where $N_m = 25$ is the horizon over which our impulse response remains negative (Section 3.2)—set independently of, and considerably longer than, N_c. Setting $N_m = N_c$ recovers their specification exactly as a special case. The key substantive point is that the labor market shock and the mortality response are governed by two different clocks. The economy recovers after $N_c = 10$ years, but the mortality benefit does not expire when the economy recovers. We write Δ_{GR}^{dm} for the welfare cost of equation (6) evaluated along this deterministic path, and compute it both with mortality held exogenous ($dm = 0$) and with mortality endogenous, at $N_m = N_c = 10$ and at $N_m = 25$, to isolate the contribution of recession-induced mortality declines and of their duration.

4.3.2 Results

Table 5 reports Δ_{GR}^{dm} as a percentage of average annual consumption. The economic shock is identical in every column, so all variation across columns comes from the mortality side. Column (1) holds mortality exogenous: the cost of the recession is positive at every age and risk aversion, ranging from 2.29 percent at age 35 under $\gamma = 1.5$ to 3.93 percent at age 65 under $\gamma = 2.5$.

Columns (2)–(4) tie the mortality response to the economic recovery, as in Finkelstein et al. [2025]. The offset is modest at younger ages—at age 35 a \$400,000 VSLY lowers the cost only from 2.29 to 2.00 percent—because a 35-year-old faces low baseline mortality over the ten years the discount applies. It grows with age as baseline mortality rises: at age 65 the same VSLY moves the cost from 2.34 to −1.40 percent, and at 75 from 1.69 to −6.27 percent.

Columns (5)–(7) extend the discount to $N_m = 25$ years while leaving the ten-year economic shock untouched. The additional fifteen years of reduced mortality roughly triples the offset at every age. At age 45 and $\gamma = 1.5$, a \$400,000 VSLY lowers the cost from 2.13 percent to 1.36 percent when mortality tracks the recovery, but to −1.68 percent once the mortality response is allowed to outlast it—the difference between a recession that is costly and one that is welfare-improving. The same comparison at age 65 moves the cost from -1.40 to -7.09 percent, and at age 75 from -6.27 to -10.71 percent. Decoupling the two clocks also lowers the age at which the recession becomes welfare-improving: under a $400,000 VSLY it falls from 65 to 45, and under $250,000 from 75 to 55.

The gain from persistence is largest in relative terms at middle ages. A 35-year-old discounts twenty-five years of slightly lower mortality heavily, and a 75-year-old already receives a large offset in the contemporaneous case, so it is agents in their forties and fifties—old enough to face non-trivial baseline mortality, young enough that the full twenty-five year horizon is relevant—whose welfare assessment changes most. The reversal in γ with age carries over from Section 4.2.2. Both features follow from the interaction of the survival function with the age profile of baseline mortality rather than from anything specific to our mortality estimates.

The central takeaway of our welfare exercise is that under a given set of shocks, or solely one, the duration of the mortality benefit from recessions has a significant impact on overall welfare costs, emphasizing the importance of our long term empirical estimates.

5 Conclusion

This paper makes important contributions to the literature on recessions and mortality. First, we use the results from independently replicating Finkelstein et al. [2025]'s Great Recession study at the commuting zone, county, and state levels to extend their event-study design to all six major U.S. recessions since 1969, estimating using a variety of window specifications. Second, we estimate local projections over a 25-year horizon to assess whether the short-run mortality decline is transient or persistent. Lastly, we incorporate our findings into existing welfare models to investigate the impact of our novel findings on the lifetime welfare cost of recessions.

The results are consistently strong across our empirical contributions. The Great Recession replication confirms the Finkelstein et al. [2025] finding, and the stacked event study shows it is not specific to the 2007–2009 episode: the same qualitative response appears across five earlier recessions that differ in origin, monetary regime, and severity, with a pooled coefficient of -0.66 against -1.02 for the Great Recession alone. The local projection impulse response remains negative through 25 years and does not reverse sign at any horizon, while the unemployment response returns to zero within a decade. Both findings are new, and together they revise the standard interpretation of the relationship between business cycles and mortality.

Turning to welfare, we show that the quantitative changes in the length and magnitude of the mortality impact of recessions that we observe empirically drastically alter the long run welfare costs of both a single recession event or a series of repeated downturns. Our results also amplify distributional effects across age, giving insight into tensions that arise across business cycles.

Several questions remain open. The heterogeneity we document raises questions about which channels drive the aggregate effect. Differential exposure to recession externalities—occupational hazard, traffic, pollution, and access to healthcare—likely varies across demographic groups in ways that our design cannot separate. The long-run persistence documented by the local projections is similarly in need of a mechanism. Cohort effects, whereby reduced pollution or traffic during a recession improves health outcomes for children in formative years, offer one candidate. Behavioral persistence and structural changes in public health infrastructure offer others. Identifying the precise mechanism is left for future work. What the evidence does establish is that the mortality benefits of recessions are not confined to the Great Recession and last much longer than previously identified. Accounting for them changes how we should evaluate the welfare costs of business cycles throughout the postwar period.

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Figures and Tables

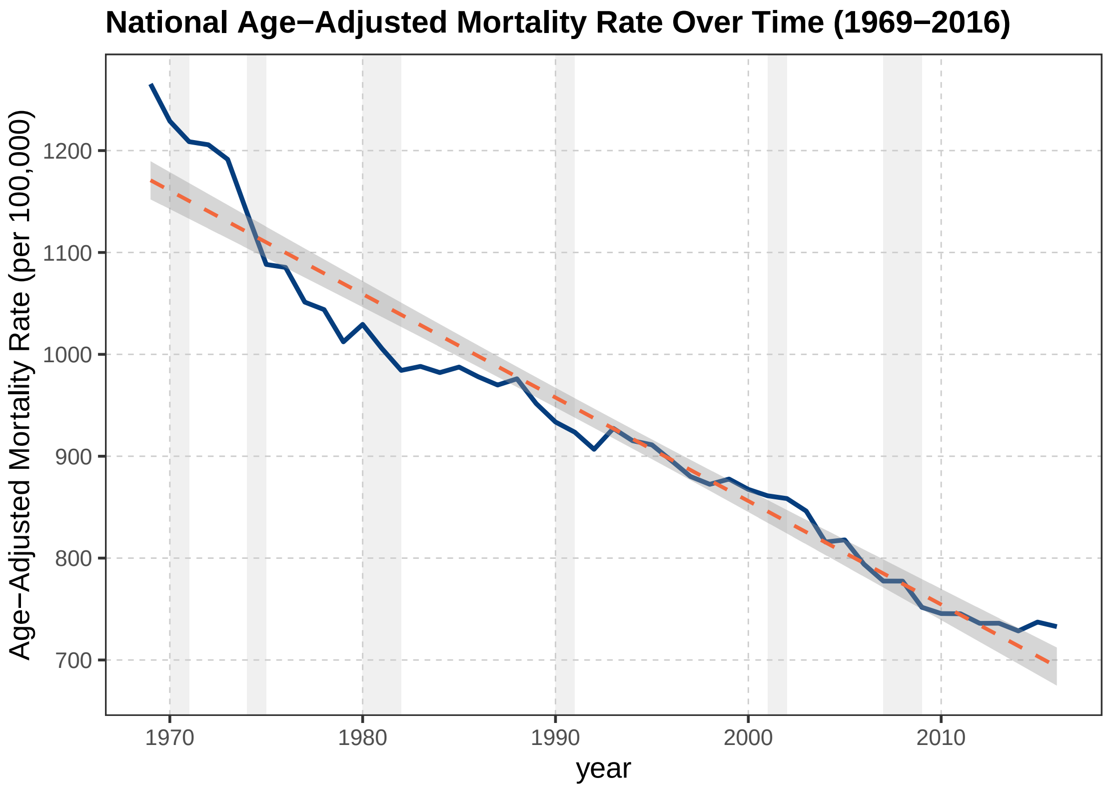


Figure 1: Age-Adjusted Mortality Rate Over Time, U.S. States (1969–2016)

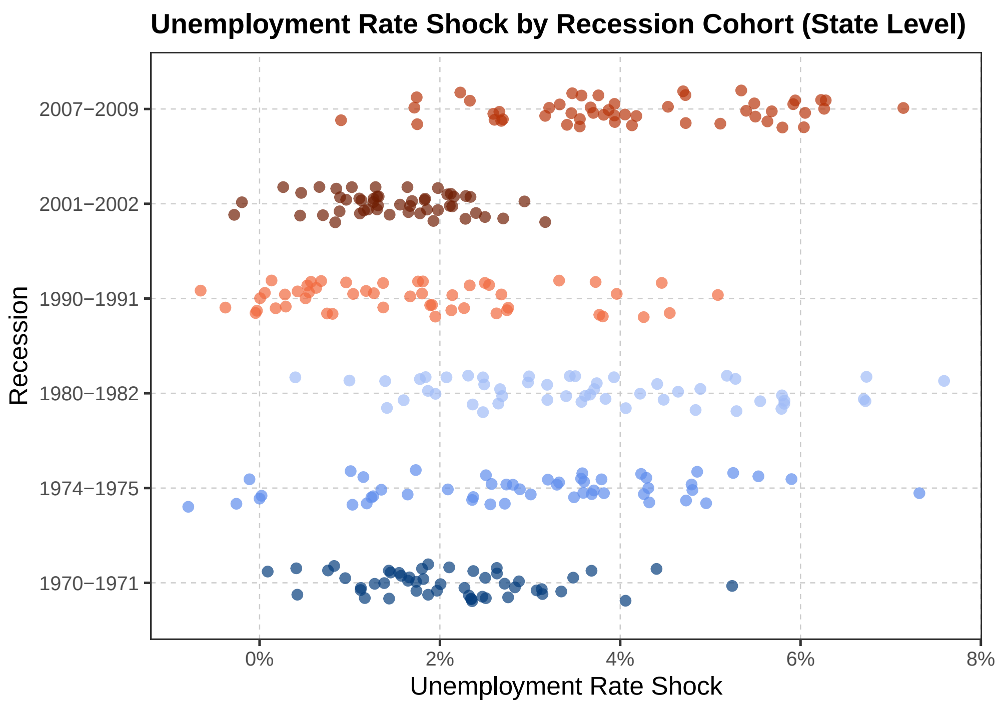
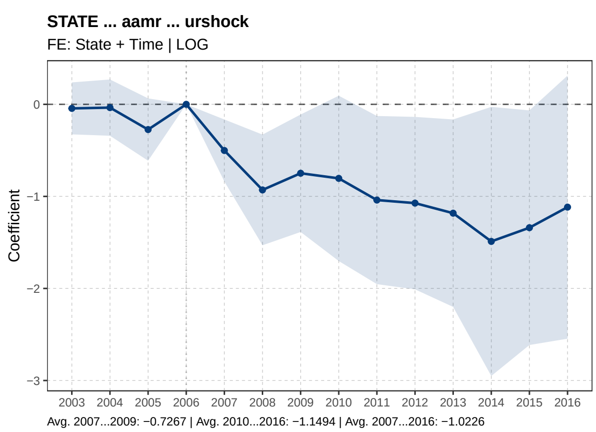
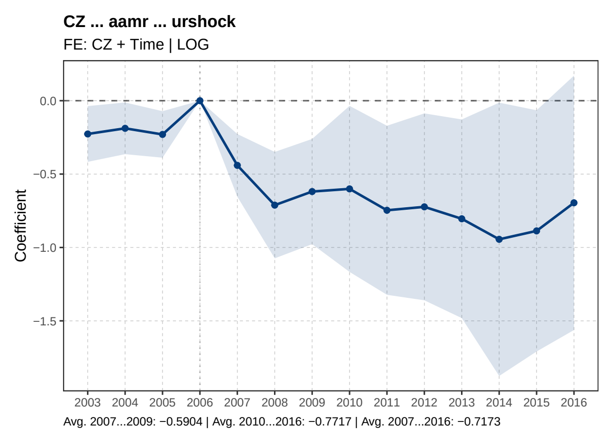


Figure 2: Unemployment Rate Shock Magnitude by Recession



(a) State Level



(b) CZ Level

Figure 3: Event Study: Effect of UR Shock on log(AAMR) (Great Recession)

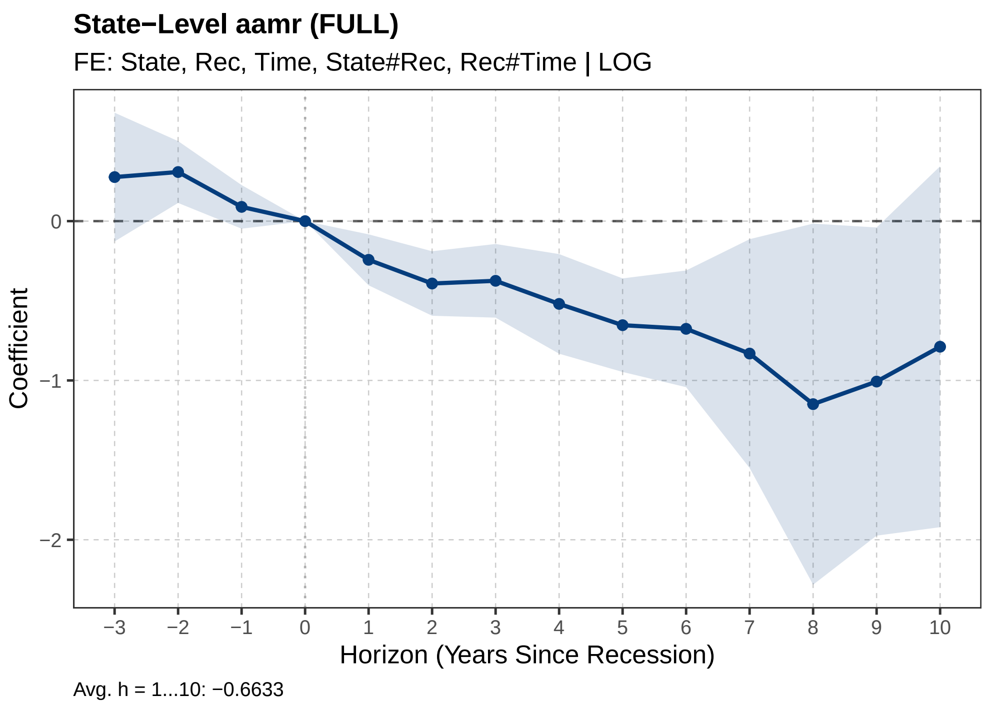


Figure 4: Stacked Event Study: Effect of UR Shock on log(AAMR) (6 Recessions, State Level)

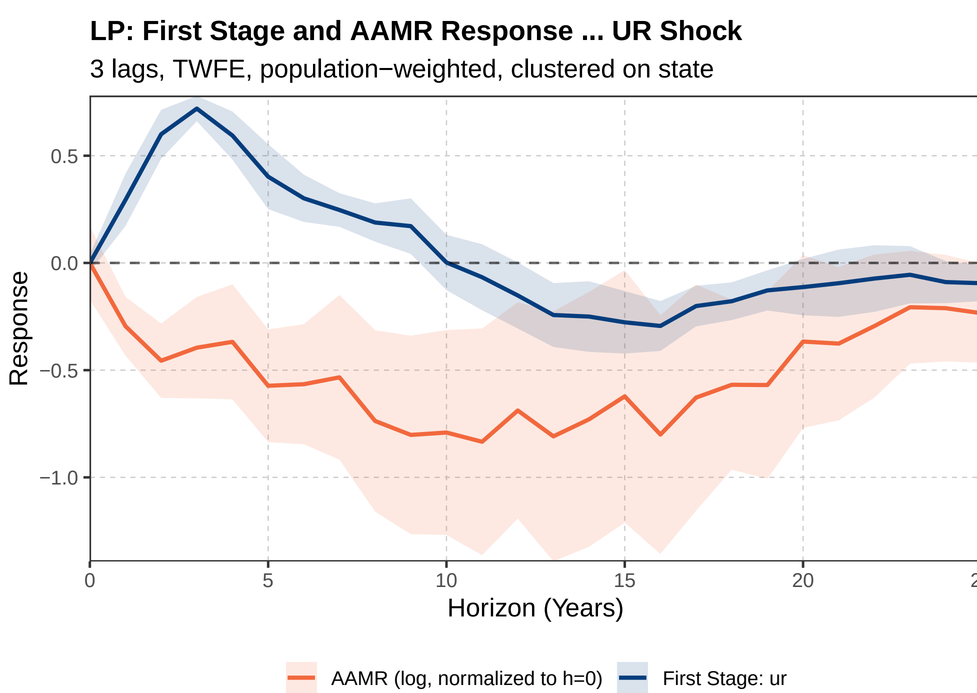
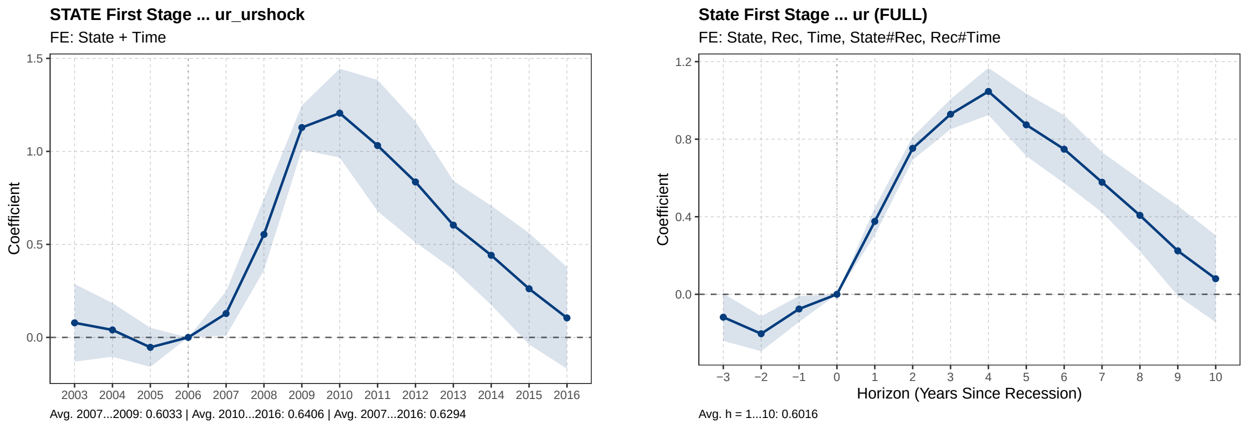


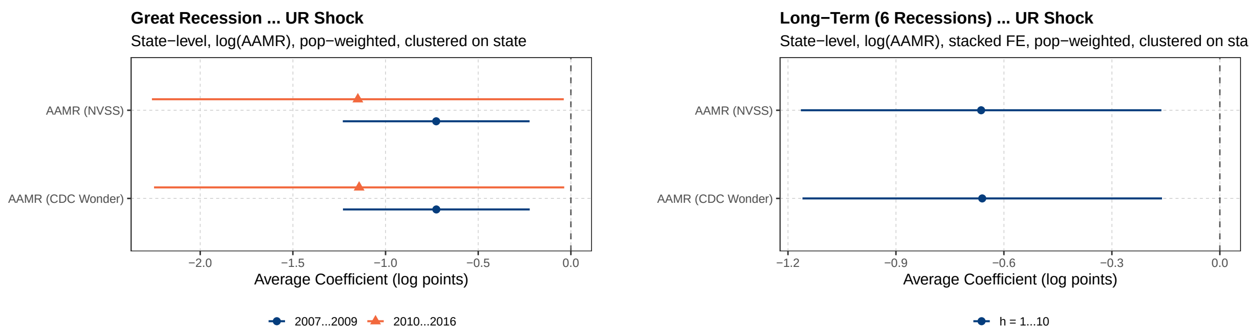
Figure 5: Local Projection: Effect of UR Shock on log(AAMR), Unemployment Rate (State Level, 25-Year Horizon)



(a) Great Recession, State Level

(b) 6 Recessions, State Level

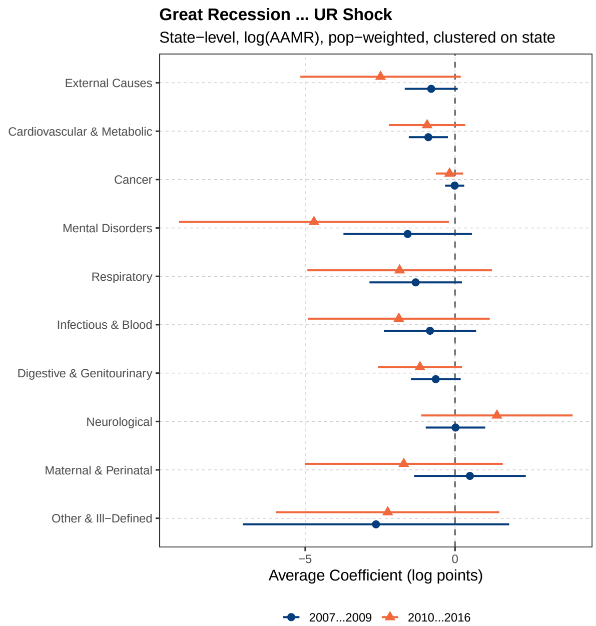
Figure 6: First Stage: UR Shock on Unemployment Rate



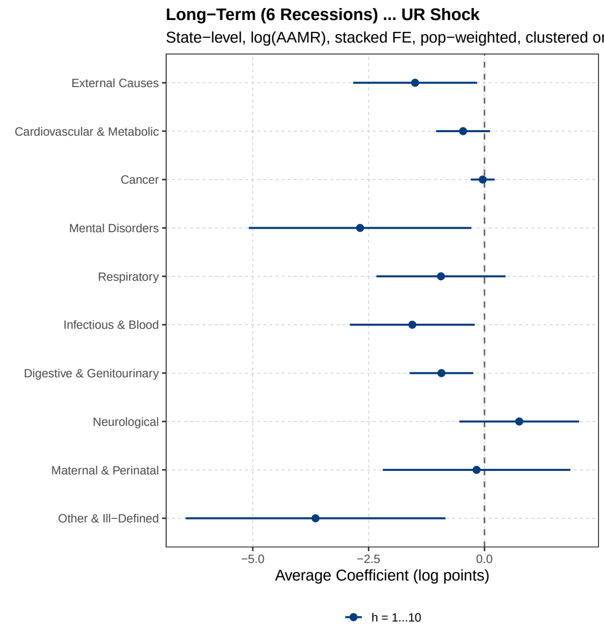
(a) Great Recession

(b) Long-Term (6 Recessions)

Figure 7: Average Coefficients: UR Shock (Overview)

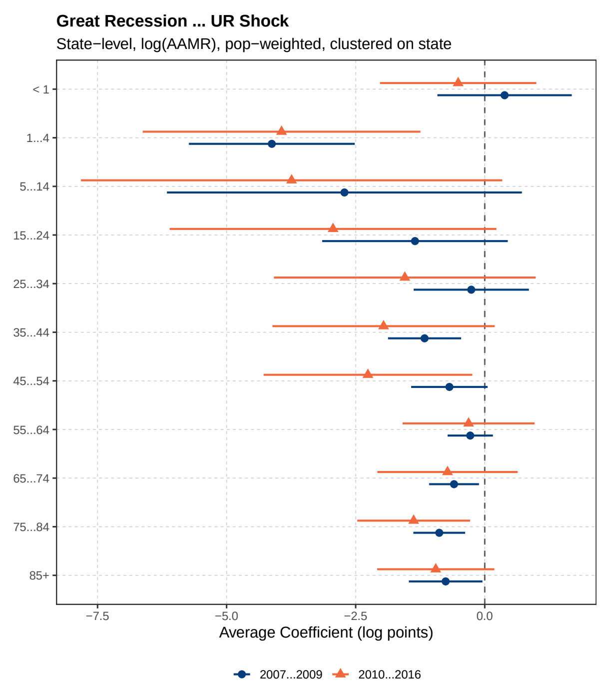


(a) Great Recession

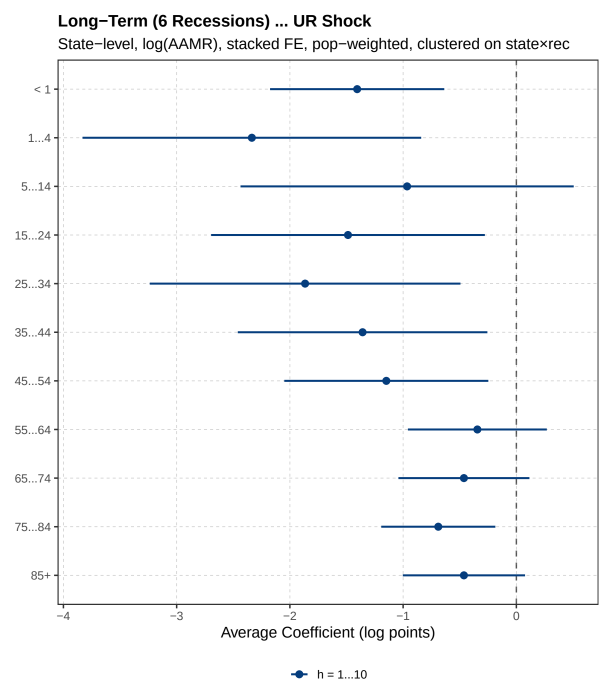


(b) 6 Recessions

Figure 8: Average Coefficients by Cause of Death (UR Shock)

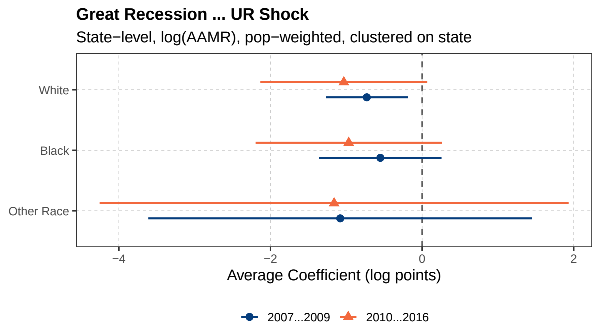


(a) Great Recession

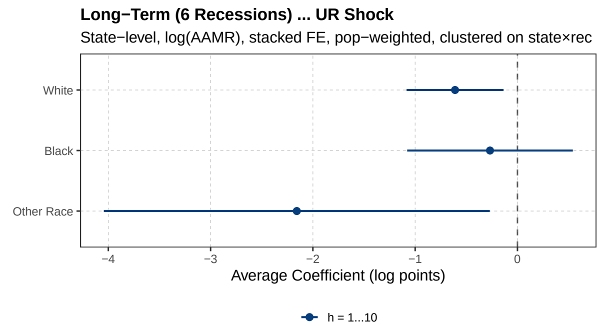


(b) 6 Recessions

Figure 9: Average Coefficients by Age Group (UR Shock)

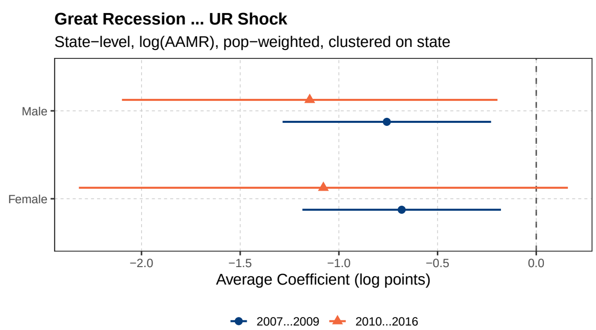


(a) Great Recession

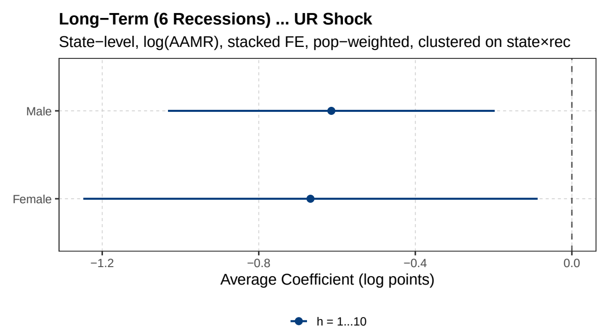


(b) Long-Term (6 Recessions)

Figure 10: Average Coefficients by Race: UR Shock



(a) Great Recession



(b) Long-Term (6 Recessions)

Figure 11: Average Coefficients by Sex: UR Shock

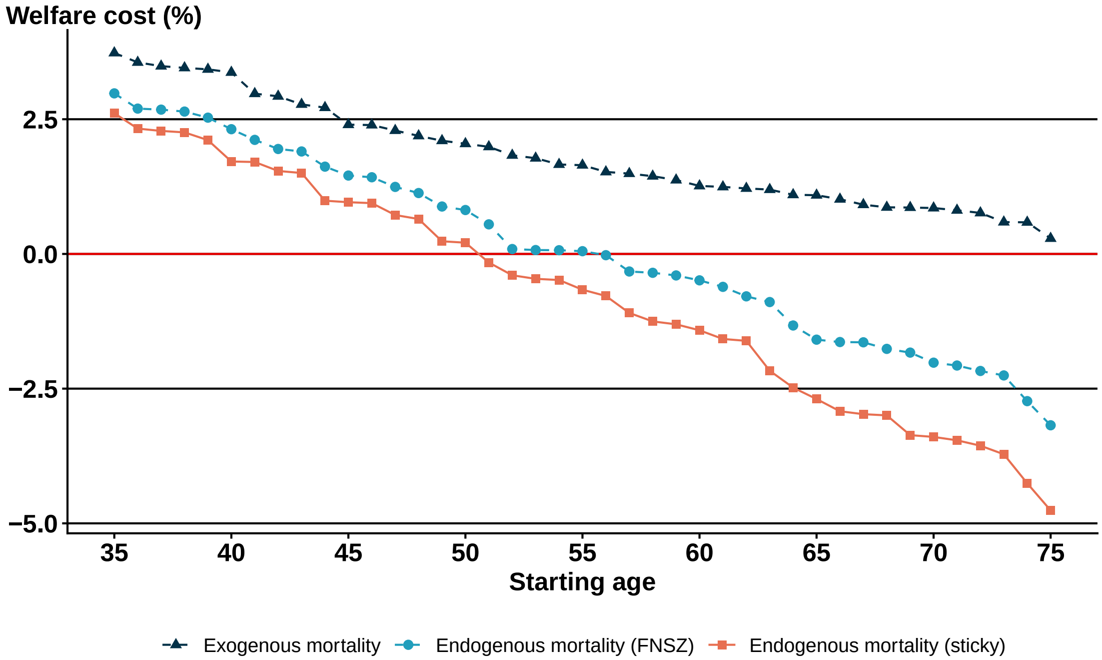


Figure 12: Welfare Cost of Business Cycles by Starting Age ($\gamma = 2$, VSLY $=$ \$250k)

Table 1: Summary Statistics

	Panel A: GR Sample (2003–2016)			Panel B: Full Panel (1969–2016)		
	Mean	SD	N	Mean	SD	N
Economic Indicators						
UR (%)	6.06	2.06	686	5.87	1.88	2601
UR Shock (pp)	4.17	1.38	686	2.74	1.55	2601
Mortality (per 100,000)						
AAMR, CDC	785.3	90.1	686	979.9	164.5	2601
AAMR, Constructed	784.8	90.4	686	979.5	164.9	2601
AAMR, White	769.6	88.5	686	955.3	155.7	2601
AAMR, Black	863.6	207.0	684	1188.9	315.2	2569
AAMR, Other	513.4	218.9	680	637.8	372.5	2576
AAMR, Male	932.4	113.4	686	1239.8	238.0	2601
AAMR, Female	667.4	74.7	686	785.2	114.1	2601

Notes: AAMR is age-adjusted mortality rate per 100,000 population. CDC AAMR uses published CDC WONDER figures; Constructed AAMR is calculated from CDC death counts and SEER population data using national age weights. UR is from BLS LAUS (Panel A) and Fieldhouse et al. [2024] (Panel B). UR Shock is the state-specific peak-to-trough change in the unemployment rate.

Table 2: Great Recession: First Stage Average Coefficients

	2007–2009	2010–2016
UR on UR Shock	0.6009***	0.6335***
	(0.0712)	(0.1360)
log(GDP) on GDP Shock	-0.5061***	-0.9999***
	(0.0791)	(0.2215)
log(GDP) on UR Shock	-1.1535***	-2.3427**
	(0.4209)	(0.9318)
UR on GDP Shock	0.1261***	0.0899
	(0.0290)	(0.0616)

Table 3: Long-Term (6 Recessions): First Stage Average Coefficients

h = 1–10
0.6016***
(0.0551)
-1.0447***
(0.1162)
-2.2799***
(0.5030)
0.1433***
(0.0289)

Table 4: Welfare Cost of Business Cycles: Contemporaneous versus Persistent Mortality

	Exogenous	Endogenous mortality					
		FNSZ			Sticky		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel (a): Starting age 35							
$\gamma = 1.5$	3.06	2.73	2.28	1.83	2.56	1.89	1.23
$\gamma = 2$	3.73	3.42	2.98	2.55	3.26	2.61	1.97
$\gamma = 2.5$	4.49	4.20	3.79	3.39	4.06	3.45	2.86
Panel (b): Starting age 45							
$\gamma = 1.5$	1.59	1.17	0.61	0.05	0.96	0.12	-0.71
$\gamma = 2$	2.10	1.69	1.13	0.57	1.49	0.65	-0.18
$\gamma = 2.5$	2.62	2.23	1.69	1.16	2.04	1.23	0.44
Panel (c): Starting age 55							
$\gamma = 1.5$	1.19	0.57	-0.20	-0.97	0.26	-0.89	-2.02
$\gamma = 2$	1.65	0.96	0.09	-0.76	0.63	-0.66	-1.92
$\gamma = 2.5$	2.11	1.38	0.43	-0.51	1.02	-0.39	-1.75
Panel (d): Starting age 65							
$\gamma = 1.5$	0.59	-0.37	-1.48	-2.56	-0.84	-2.47	-4.05
$\gamma = 2$	0.86	-0.34	-1.76	-3.14	-0.92	-3.00	-4.98
$\gamma = 2.5$	1.14	-0.34	-2.12	-3.82	-1.05	-3.62	-6.03
VSLY	-	\$100k	\$250k	\$400k	\$100k	\$250k	\$400k

Notes: Each cell reports the welfare cost of business cycles Δ^{dm}, expressed as a percentage of average annual consumption, for an agent of the indicated starting age and coefficient of relative risk aversion γ. Column (1) holds mortality exogenous. Columns (2)–(4) follow Finkelstein et al. [2025] in applying the mortality discount only while the aggregate state is bad. Columns (5)–(7) apply the persistent specification of equation (8), in which the discount also applies for one period after the aggregate state recovers. Both endogenous specifications use the same dm. The value of a statistical life-year (VSLY) is reported in the final row. Negative values indicate that recession risk is welfare-improving. The income process follows Krebs [2007]; remaining parameters follow Finkelstein et al. [2025].

Table 5: Welfare Cost of the Great Recession: Mortality Response Tied to versus Decoupled from the Economic Recovery

		Endogenous mortality					
	Exogenous	$N_m = N_c = 10$ (FNSZ)			$N_m = 25$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel (a): Starting age 35							
$\gamma = 1.5$	2.29	2.22	2.11	2.00	1.80	1.13	0.47
$\gamma = 2$	3.19	3.12	3.01	2.91	2.74	2.10	1.46
$\gamma = 2.5$	4.21	4.15	4.05	3.96	3.82	3.23	2.65
Panel (b): Starting age 45							
$\gamma = 1.5$	2.13	1.92	1.64	1.36	1.02	-0.35	-1.68
$\gamma = 2$	2.96	2.72	2.41	2.10	1.76	0.25	-1.22
$\gamma = 2.5$	3.89	3.64	3.31	2.97	2.64	1.03	-0.51
Panel (c): Starting age 55							
$\gamma = 1.5$	1.67	1.19	0.60	0.02	-0.32	-2.61	-4.82
$\gamma = 2$	2.42	1.85	1.14	0.44	0.11	-2.62	-5.20
$\gamma = 2.5$	3.26	2.61	1.77	0.95	0.65	-2.48	-5.38
Panel (d): Starting age 65							
$\gamma = 1.5$	2.34	1.20	-0.11	-1.40	-0.65	-3.95	-7.09
$\gamma = 2$	3.10	1.67	-0.03	-1.68	-0.60	-4.75	-8.56
$\gamma = 2.5$	3.93	2.16	0.03	-1.99	-0.58	-5.58	-10.00
Panel (e): Starting age 75							
$\gamma = 1.5$	1.69	-0.83	-3.61	-6.27	-2.36	-6.67	-10.71
$\gamma = 2$	2.23	-1.12	-4.85	-8.30	-3.08	-8.72	-13.74
$\gamma = 2.5$	2.82	-1.56	-6.37	-10.63	-4.02	-11.06	-16.93
VSLY	-	\$100k	\$250k	\$400k	\$100k	\$250k	\$400k

Notes: Each cell reports the welfare cost of the Great Recession Δ^{dm}_{GR}, expressed as a percentage of average annual consumption, for an agent of the indicated starting age and coefficient of relative risk aversion γ. The income process follows Krebs [2007]; baseline mortality is taken from the 2007 Social Security Administration period life table; remaining parameters follow Finkelstein et al. [2025].

A Data

Mortality. We obtain death counts and age-adjusted mortality rates from the CDC WONDER Compressed Mortality Files [see Centers for Disease Control and Prevention, 2024b], which span three reporting eras: 1968–1978, 1979–1998, and 1999–2016. We append all three and restrict to 1969–2016. For our primary outcome, we use the CDC-provided age-adjusted mortality rate (AAMR) per 100,000 population, which applies the 2000 U.S. Standard Population weights [Centers for Disease Control and Prevention, 2024a]. As a robustness check, we construct our own AAMR by combining age-specific death counts from WONDER with single-year-of-age population data from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program [National Cancer Institute, 2024] and applying the same 2000 Standard Age Weights. For heterogeneity analyses, we construct race-specific AAMR (White, Black, Other), sex-specific AAMR (Male, Female), age-specific crude mortality rates for eleven age bins (<1, 1–4, 5–14, 15–24, 25–34, 35–44, 45–54, 55–64, 65–74, 75–84, 85+), and cause-of-death AAMR for ten aggregated ICD categories (external causes, cardiovascular and metabolic, cancer, mental disorders, respiratory, infectious and blood, digestive and genitourinary, neurological, maternal and perinatal, and other/ill-defined).

Economic indicators. For the Great Recession analysis (2003–2016), we obtain annual unemployment rates from the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS) program [Bureau of Labor Statistics, 2024] at the state, county, and commuting zone levels. We construct employment-to-population (EPOP) ratios as (Labor Force − Unemployed) from LAUS divided by working-age population (16+) from SEER [National Cancer Institute, 2024]. For the full panel (1969–2016), state-level unemployment rates for 1969–1976 come from the claims-based predicted series of Fieldhouse et al. [2024], which we splice with BLS state-level aggregate unemployment rates for 1977–2016. Real GDP per capita for all years 1969–2016 is constructed from state-level GDP data from the Bureau of Economic Analysis [Bureau of Economic Analysis, 2024], deflated using state-specific GDP deflators from the replication package of Kleinman et al. [2023] and divided by SEER state population.

Shock construction. The unemployment rate shock for area c in recession r is the peak-to-trough change in the unemployment rate: $SHOCK_c^r = UR_{c,\text{trough}} - UR_{c,\text{peak}}$. For the Great Recession, this is the 2009 minus 2007 annual average unemployment rate. For the five earlier recessions, the peak and trough years are recession-specific (see Table 8). The GDP shock is constructed analogously as the change in log real GDP per capita between the same peak and trough years, multiplied by -1 so that a positive value indicates a larger negative GDP shock.

Geography. State-level analyses use the 50 states plus the District of Columbia (51 units). Commuting zones are constructed by aggregating county-level data using the 1990 county-to-CZ crosswalk from Dorn [2009], with county-level variables weighted by county population when aggregating to the CZ level. This yields 741 CZs covering the contiguous United States. County-level analyses use individual counties directly. Because the CZ and county analyses are restricted to the Great Recession period (2003–2016), county FIPS harmonization is minimal; the large majority of county boundary and code changes occurred before 2000.

Weights. For the Great Recession analysis, regressions are weighted by 2006 population (the last pre-recession year). For the full panel, we weight by each state's average population over 1969–2016 rather than by recession-specific reference-year population; the latter would implicitly upweight later recessions due to secular population growth.

Baseline survival. The welfare model requires an age-specific mortality path in normal times, $m^H(t)$, which is not estimated from our panel but taken directly from the 2007 Social Security Administration period life table [Social Security Administration, 2010]. The table reports death probabilities separately by sex; we average the male and female rates at each single year of age to obtain $m^H(t)$, and cumulative survival to age t is the running product of $1 - m^H$ over preceding ages. We use the 2007 table, and average across sexes, to match the calibration of Finkelstein et al. [2025].

B Model Derivations

This appendix reproduces the components of the welfare model that we adopt unchanged from Krebs [2007] and Finkelstein et al. [2025], so that our calibration can be replicated, and derives the one component that is our own.

B.1 Income Process

We follow Krebs [2007] exactly, as adopted by Finkelstein et al. [2025]. The aggregate state $\omega \in \{L, H\}$ is drawn each period, with the normal state $\omega = H$ drawn with probability π_H. Income in $t = 0$ is normalized to one and evolves as

$$y_{t+1} = (1 + g)(1 + \theta_{t+1})(1 + \eta_{t+1})y_t \tag{11}$$

where g is an exogenous growth rate independent of the aggregate state. The shock θ_{t+1} is independent of the aggregate state, with $\log(1 + \theta) \sim N(-\sigma^2/2, \sigma^2)$. The shock η_{t+1}

represents job displacement and does depend on the aggregate state:

$$\eta_{t+1} = \begin{cases} -d_\omega & \text{with probability } p_\omega \\ \dfrac{p_\omega d_\omega}{1-p_\omega} & \text{with probability } 1-p_\omega \end{cases} \tag{12}$$

Here p_ω is the job separation rate and d_ω the average earnings loss conditional on displacement, with $p_L > p_H$ and $d_L > d_H$. Because the agent consumes all income each period, any change in income passes one-for-one into consumption.

B.2 Calibration

We take the income process parameters $(g, \sigma, \pi_H, p_H, p_L, d_H, d_L)$ from Krebs [2007] and the remaining parameters from Finkelstein et al. [2025]. For mortality in normal times, $m^H(t)$, we use the 2007 SSA period life tables, computing a unisex rate as the population-weighted average of the male and female age-specific rates. We report results for VSLY values of \$100,000, \$250,000, and \$400,000, spanning the range discussed in Finkelstein et al. [2025], and for $\gamma \in \{1.5, 2, 2.5\}$. Given a target VSLY at reference consumption c, the level parameter b in equation (5) is set by inverting the VSLY expression:

$$b = \text{VSLY} \cdot c^{-\gamma} - \frac{c^{1-\gamma}}{1-\gamma} \tag{13}$$

B.3 The Persistence Adjustment

Finkelstein et al. [2025] show, in a simplified version of the model in which the aggregate state is drawn once at $t = 0$ and $p_H = 0$, that a first-order approximation to the welfare cost of a recession with endogenous mortality is

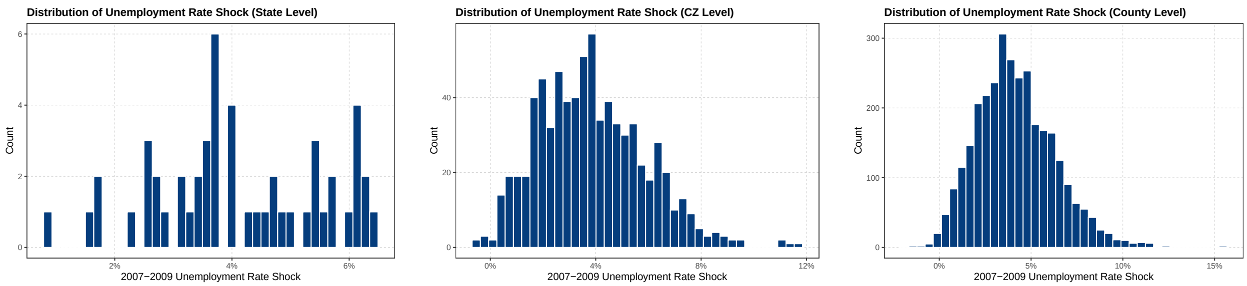
$$\Delta^{dT} \approx \Delta - dT \cdot \frac{\text{VSLY}}{c} \tag{14}$$

where Δ is the welfare cost with exogenous mortality and dT is the percentage increase in life expectancy generated by the recession. The welfare offset from reduced mortality is the increase in life expectancy times its value per year, as a share of annual consumption.

Equation (14) makes the role of persistence transparent. Nothing in it restricts how long the mortality discount applies; the duration enters only through dT. If a mortality discount dm applies for N_m periods rather than for the N_c periods the economy spends in the recession state, then dT scales with N_m, and so does the welfare offset. Our departures from Finkelstein et al. [2025] therefore do not change the structure of the welfare calculation.

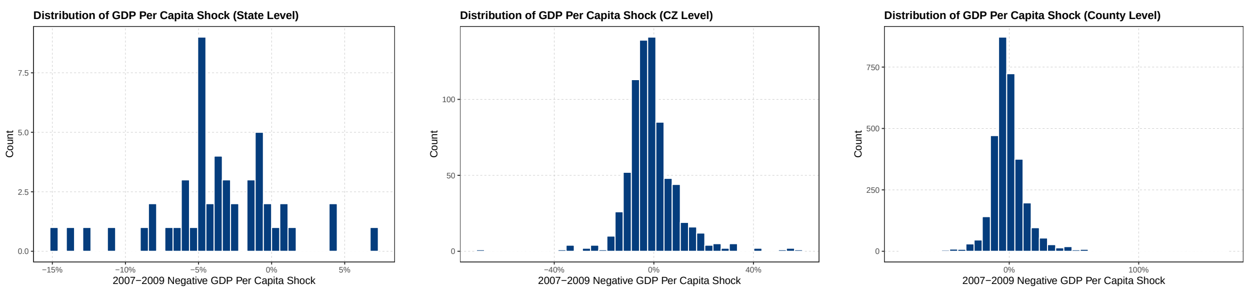
They change N_m: to $N_c + 1$ in the stochastic model of Section 4.2, through the sticky state $\tilde{L}_t$ in equation (7), and to $N_m = 25 > N_c = 10$ in the Great Recession exercise of Section 4.3. Both leave the consumption side of the model untouched, so the entire difference from their estimates operates through the second term of equation (14).

C Figures and Tables



(a) State (b) CZ (c) County

Figure 13: Distribution of UR Shocks (Great Recession)



(a) State (b) CZ (c) County

Figure 14: Distribution of GDP Shocks (Great Recession)

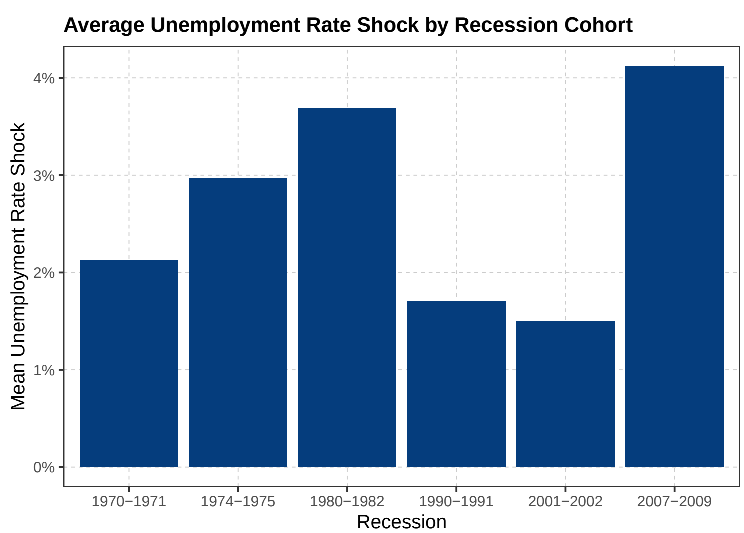
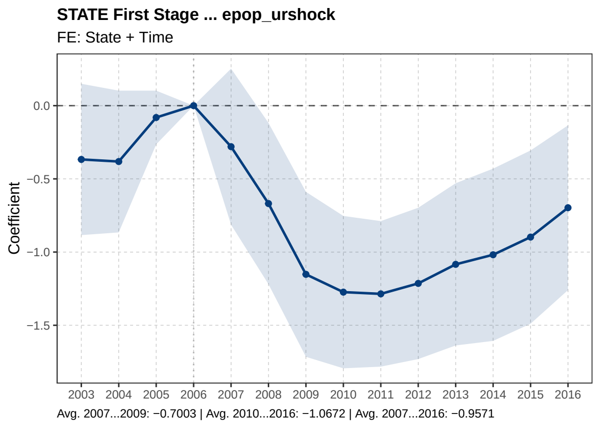
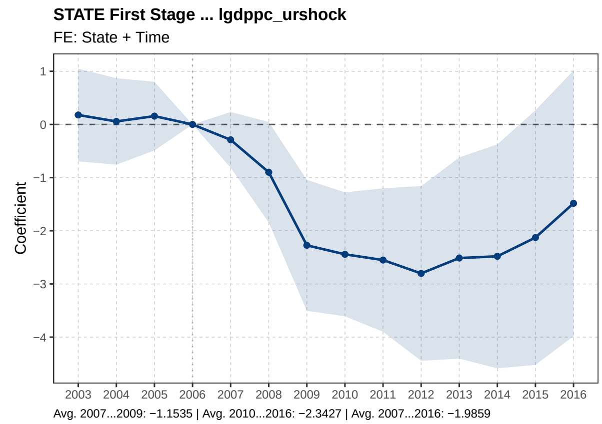


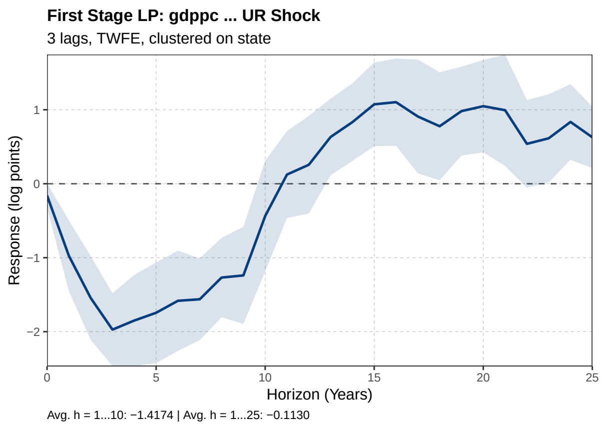
Figure 15: UR Shock Distribution Across Recessions



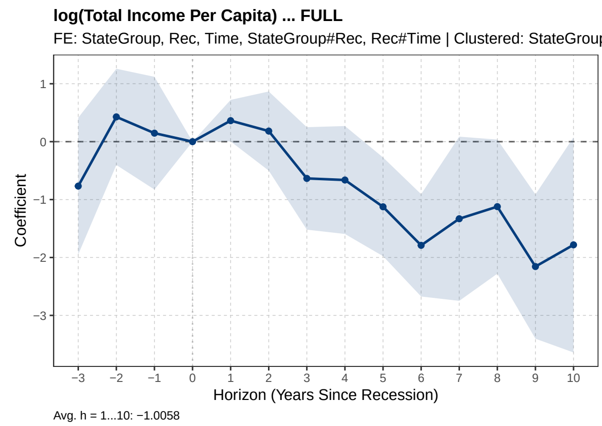
(a) EPOP ratio, Great Recession



(b) log(GDP per capita), Great Recession

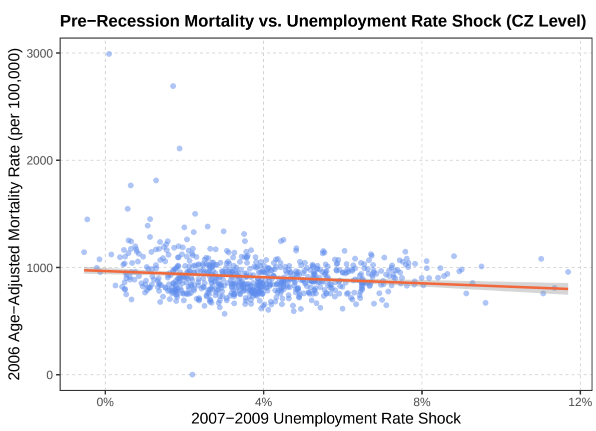


(c) log(GDP per capita), 6 recessions

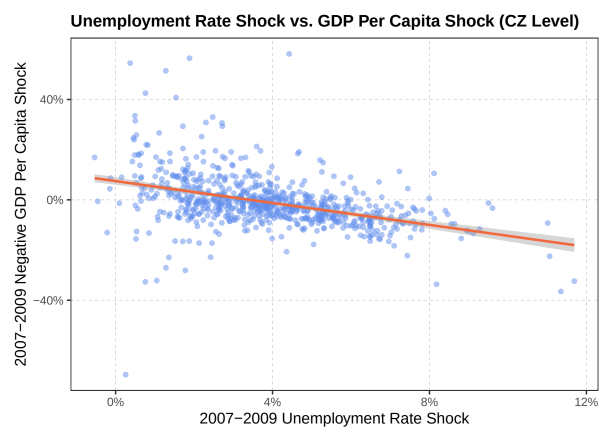


(d) log(total income per capita), 6 recessions

Figure 16: UR Shock Validation: Response of Other Economic Outcomes (State Level)



(a) Pre-Recession AAMR vs. Shock Magnitude



(b) UR Shock vs. GDP Shock Correlation

Figure 17: Descriptive Relationships (Great Recession)

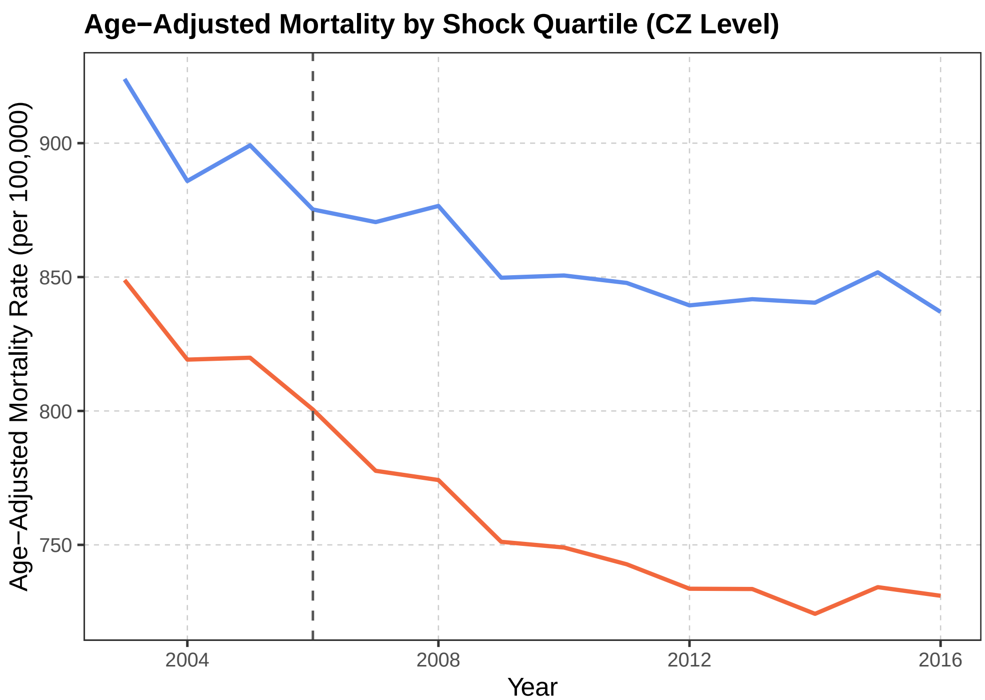


Figure 18: AAMR by Shock Quartile (Great Recession)

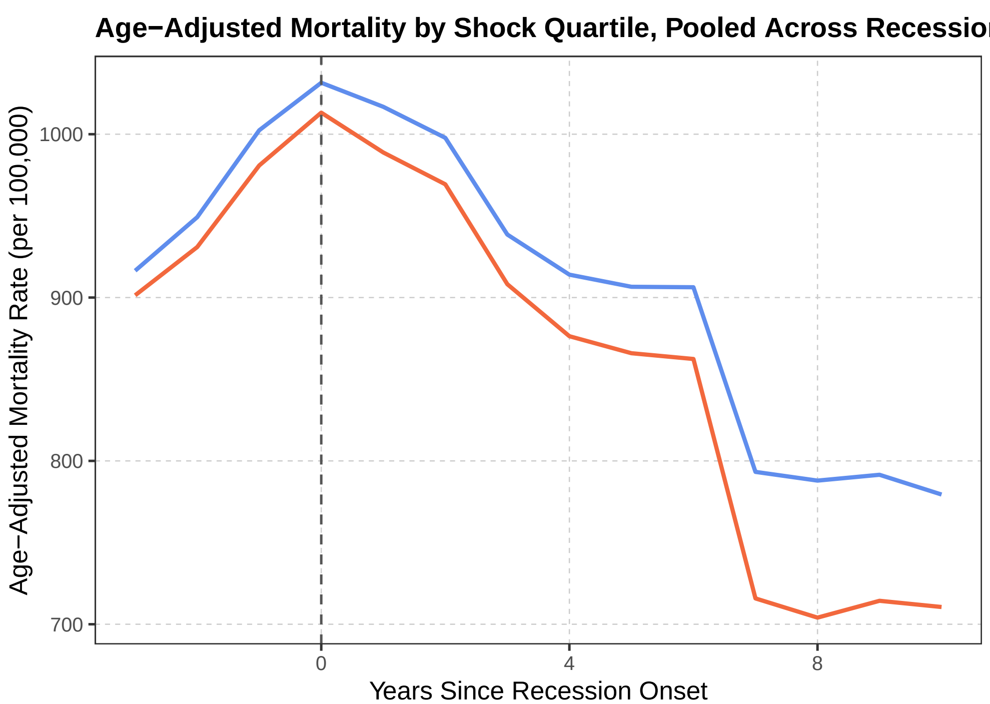


Figure 19: AAMR by Shock Quartile (Long Panel)

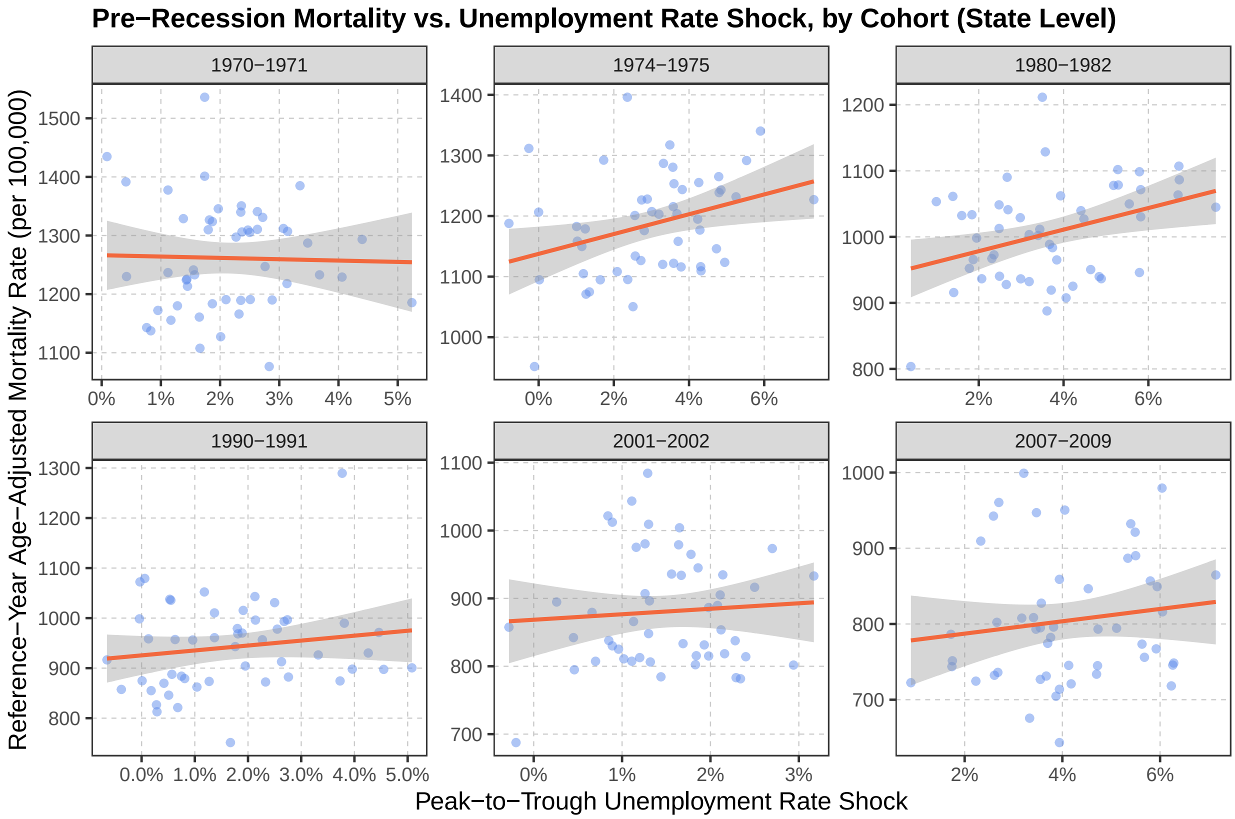
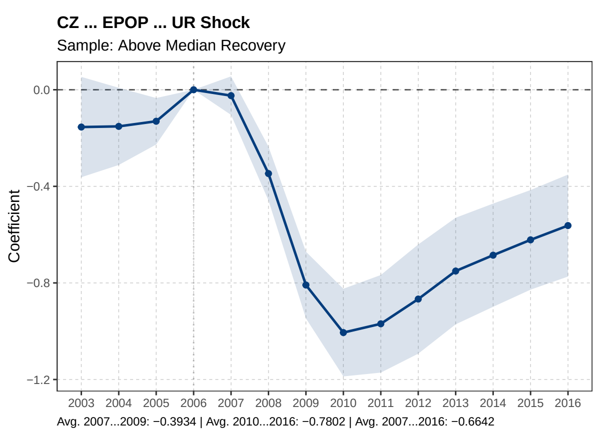
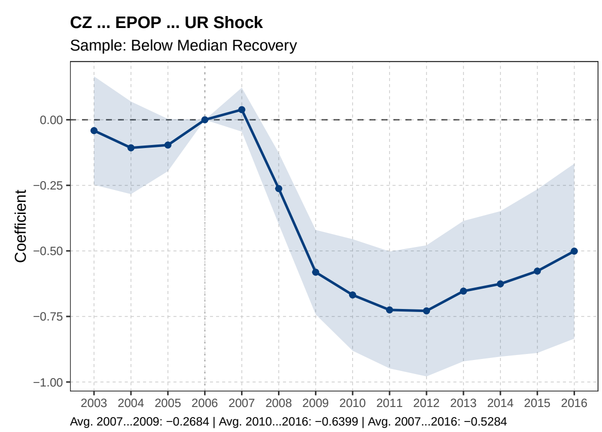


Figure 20: Pre-Recession AAMR vs. Shock Magnitude, by Recession Cohort (State Level)

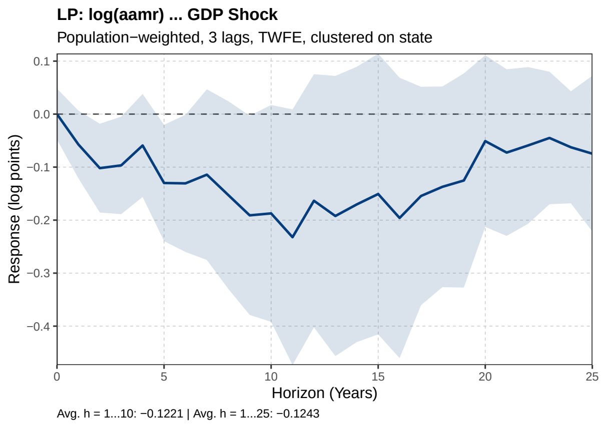


(a) Above Median Recovery

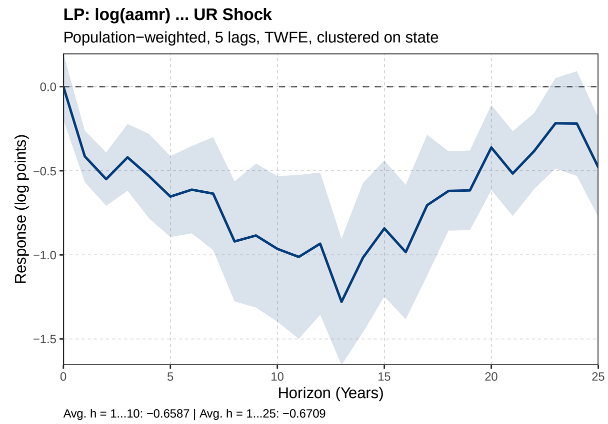


(b) Below Median Recovery

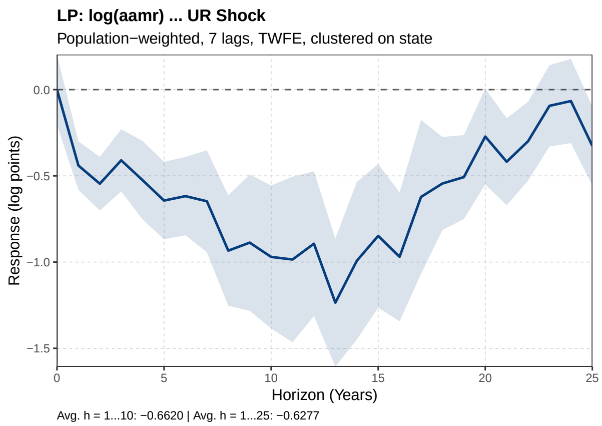
Figure 21: Effect of UR Shock on EPOP by Recovery Status (CZ Level)



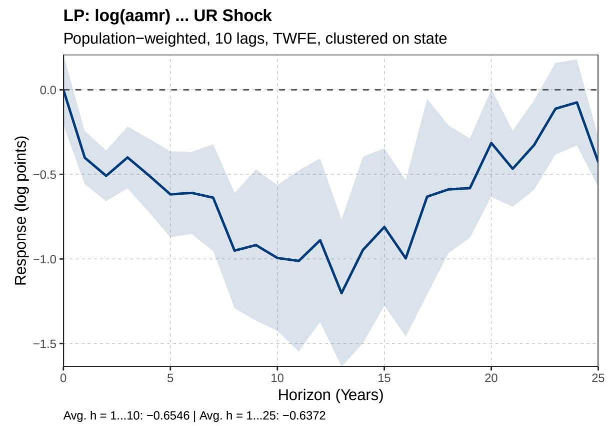
(a) GDP Shock



(b) UR Shock, 5 Lags

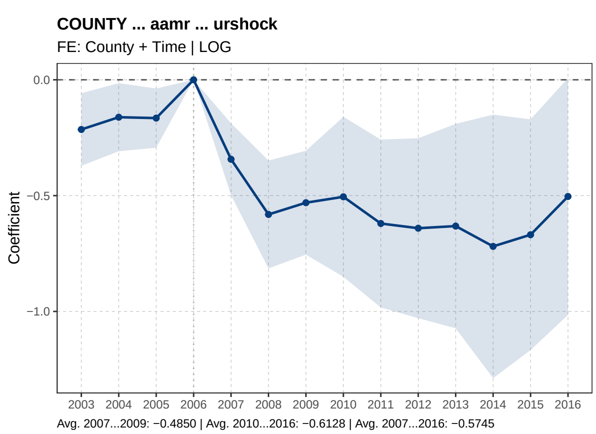


(c) UR Shock, 7 Lags

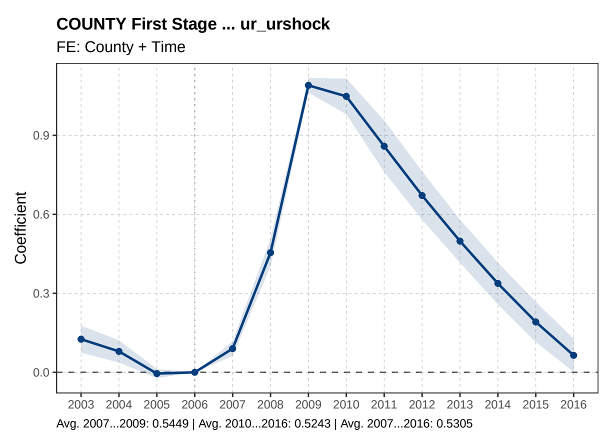


(d) UR Shock, 10 Lags

Figure 22: Local Projections: Lag Sensitivity and GDP Shock

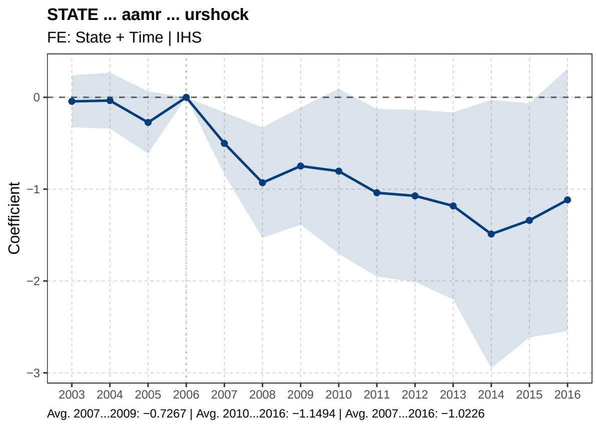


(a) County AAMR

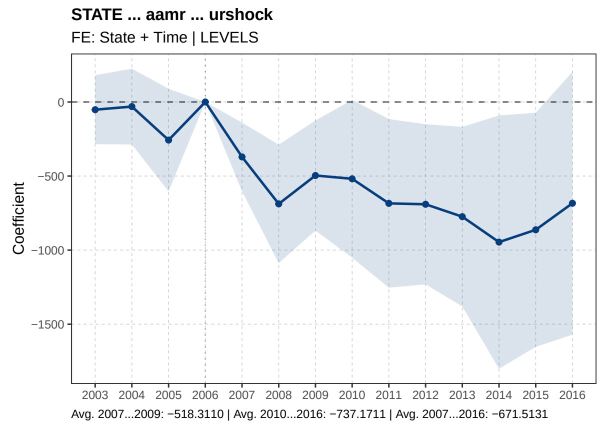


(b) County First Stage

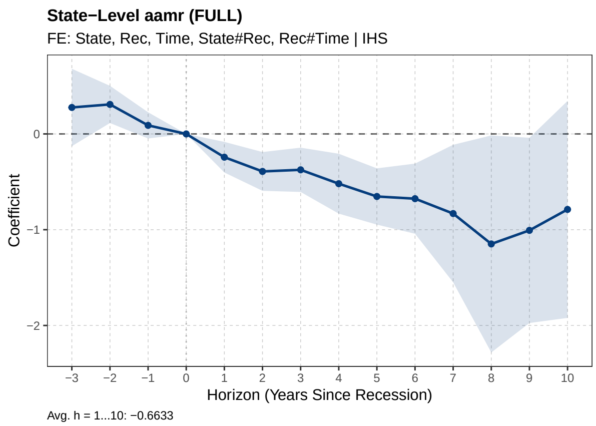
Figure 23: Great Recession at the County Level



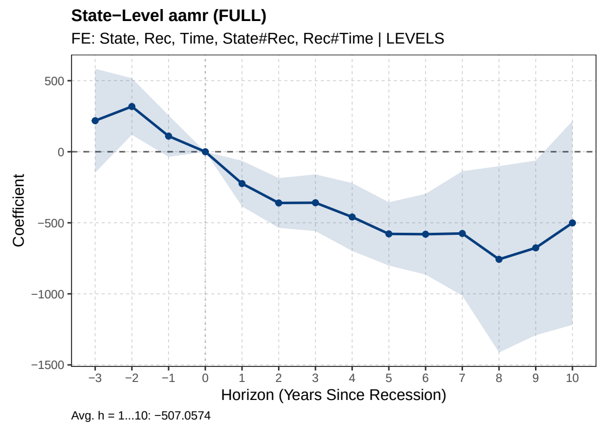
(a) IHS



(b) Levels

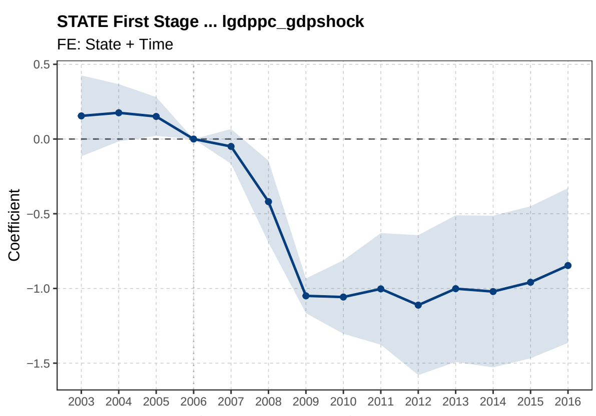


(c) IHS

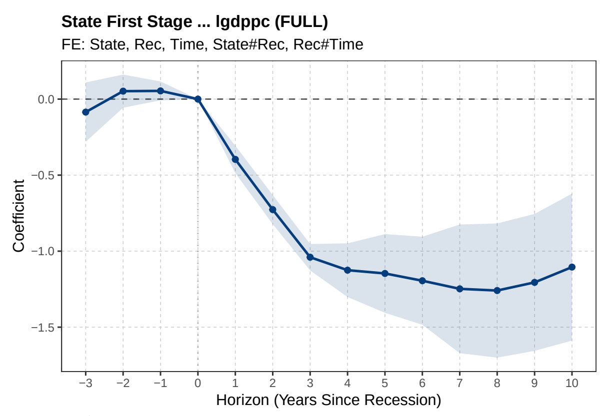


(d) Levels

Figure 24: Stacked Event Study: UR Shock on AAMR, Alternative Functional Forms (6 Recessions, State Level)

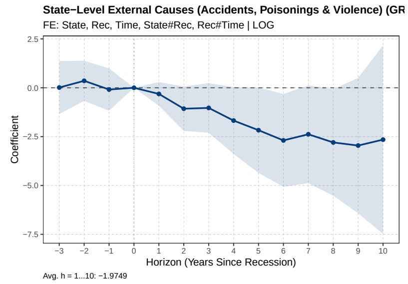


(a) Great Recession

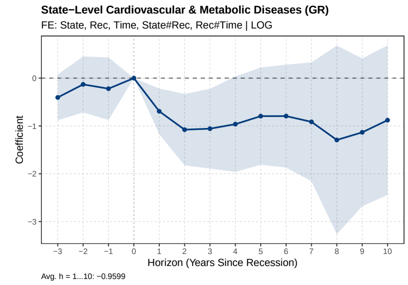


(b) 6 Recessions

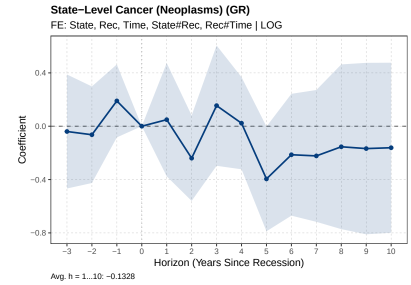
Figure 25: First Stage: GDP Shock on log(GDP per capita) (State Level)



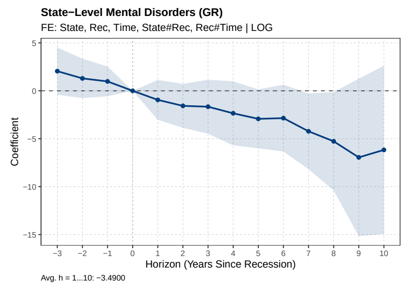
(a) External Causes



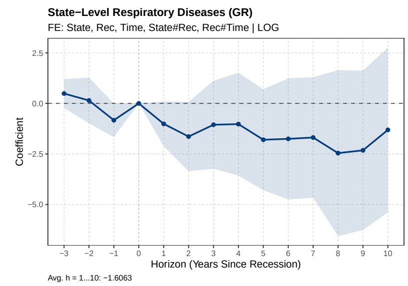
(b) Cardiovascular & Metabolic



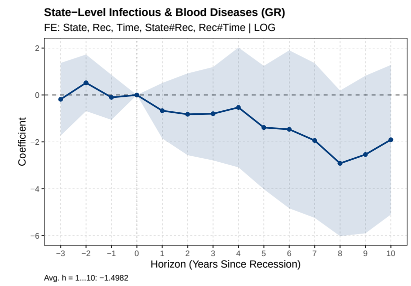
(c) Cancer



(d) Mental Disorders

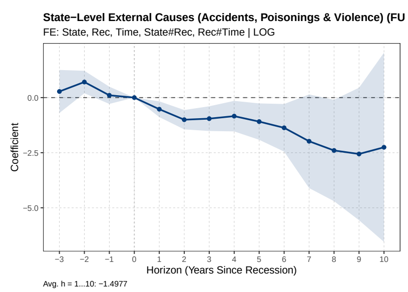


(e) Respiratory

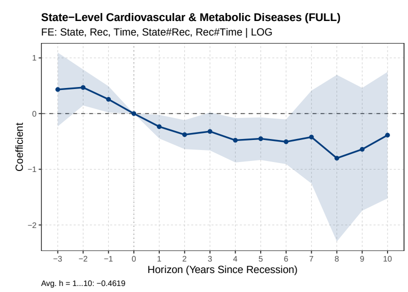


(f) Infectious & Blood

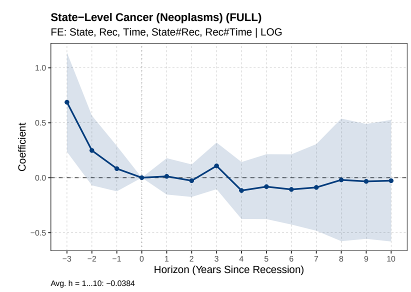
Figure 26: GR Event Studies by Cause of Death (UR Shock, State Level)



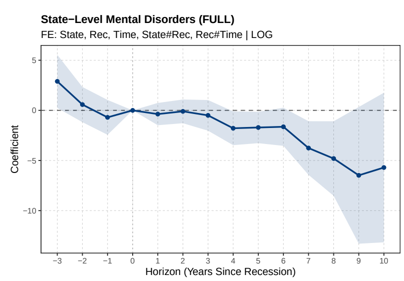
(a) External Causes



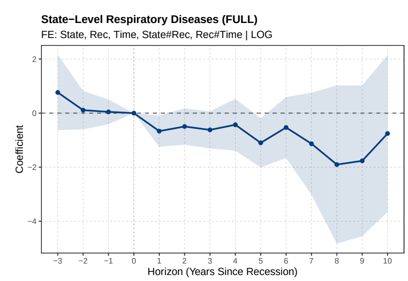
(b) Cardiovascular & Metabolic



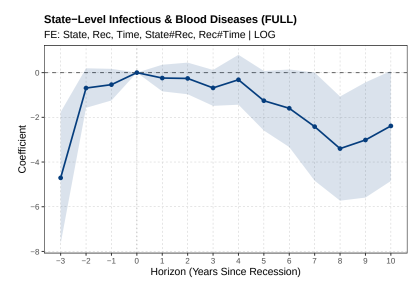
(c) Cancer



(d) Mental Disorders

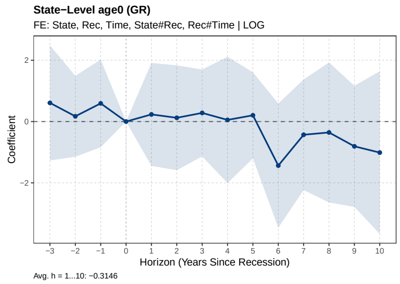


(e) Respiratory

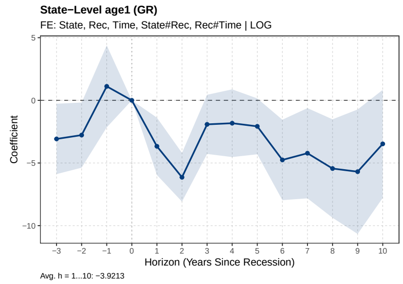


(f) Infectious & Blood

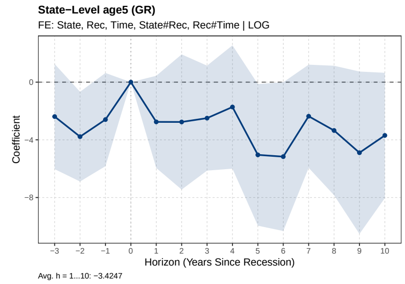
Figure 27: Full Panel Event Studies by Cause of Death (UR Shock, State Level)



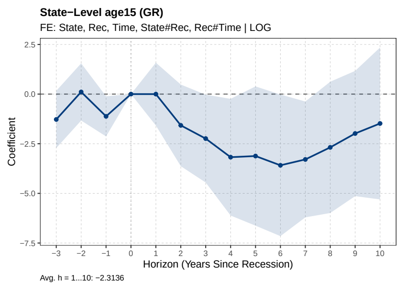
(a) Less than 1



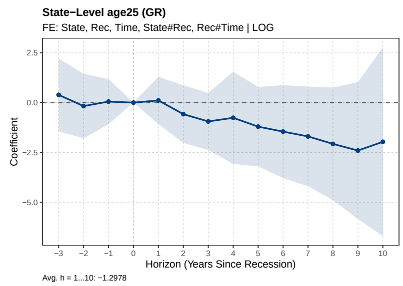
(b) 1 to 4



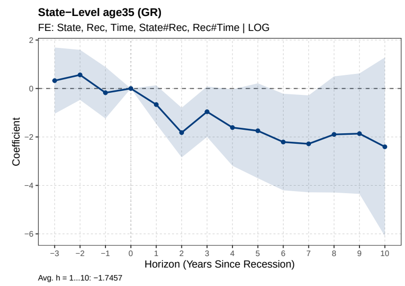
(c) 5 to 14



(d) 15 to 24

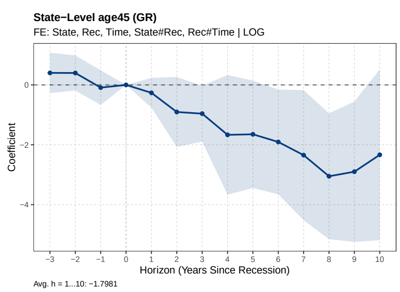


(e) 25 to 34

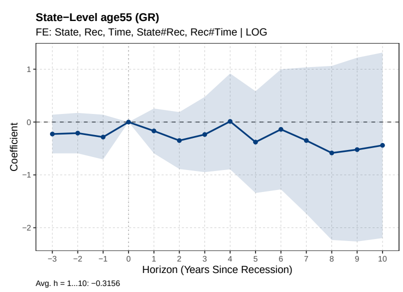


(f) 35 to 44

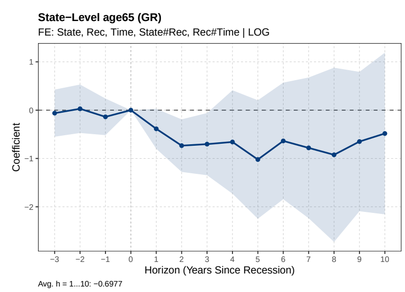
Figure 28: GR Event Studies by Age (UR Shock, State Level)



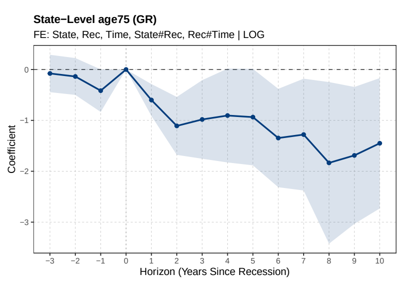
(a) 45 to 54



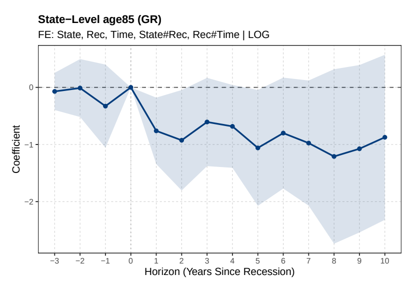
(b) 55 to 64



(c) 65 to 74

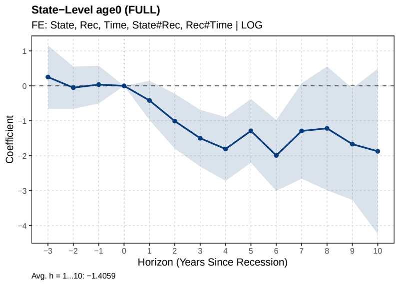


(d) 75 to 84

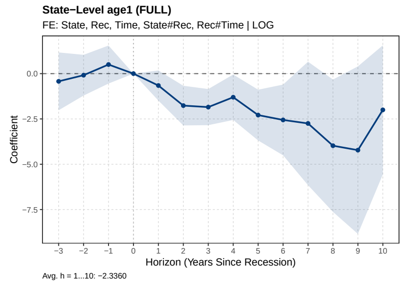


(e) 85 and over

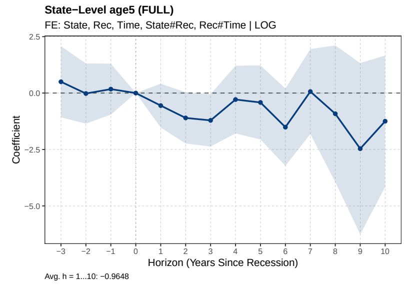
Figure 29: GR Event Studies by Age (UR Shock, State Level)



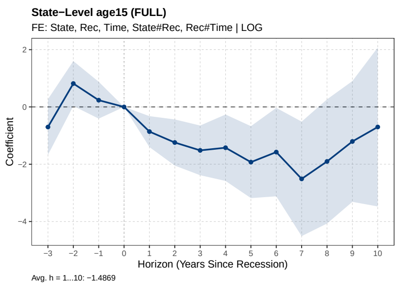
(a) Less than 1



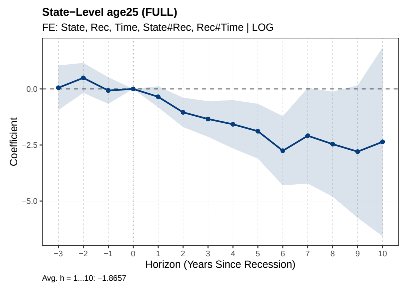
(b) 1 to 4



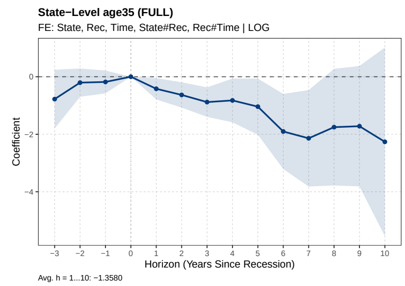
(c) 5 to 14



(d) 15 to 24

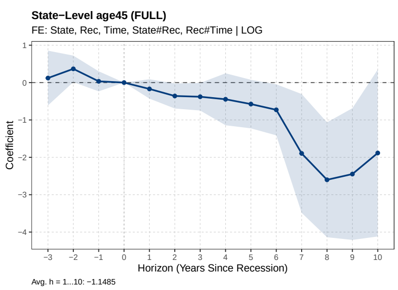


(e) 25 to 34

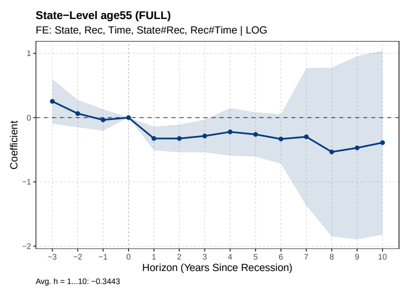


(f) 35 to 44

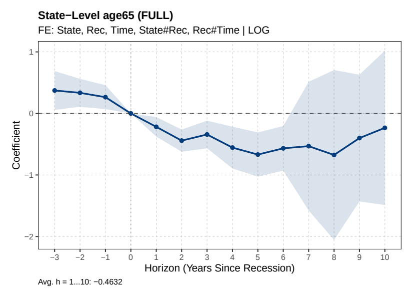
Figure 30: Full Panel Event Studies by Age (UR Shock, State Level)



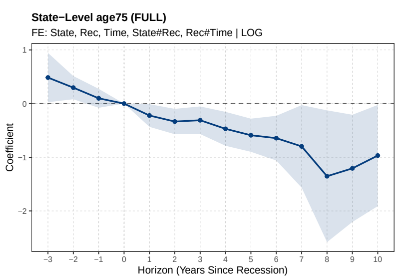
(a) 45 to 54



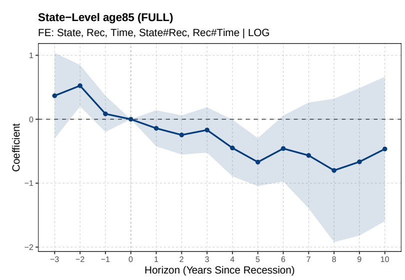
(b) 55 to 64



(c) 65 to 74

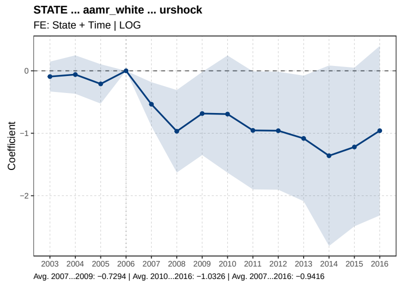


(d) 75 to 84

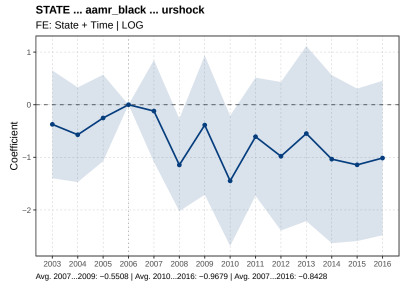


(e) 85 and over

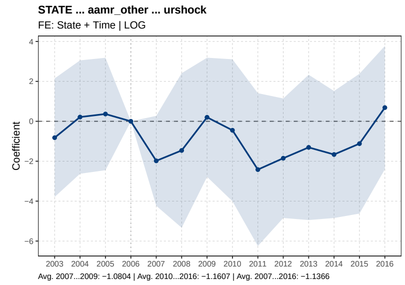
Figure 31: Full Panel Event Studies by Age (UR Shock, State Level)



(a) White

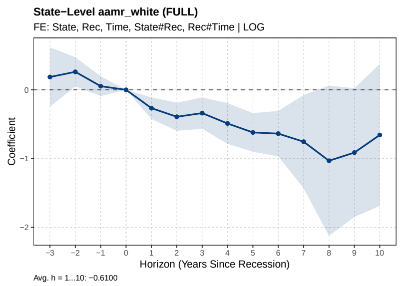


(b) Black

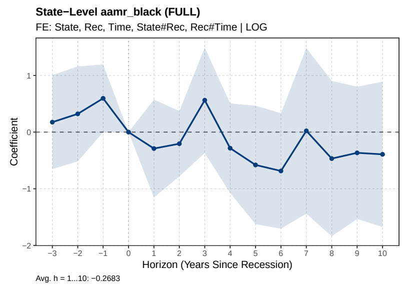


(c) Other

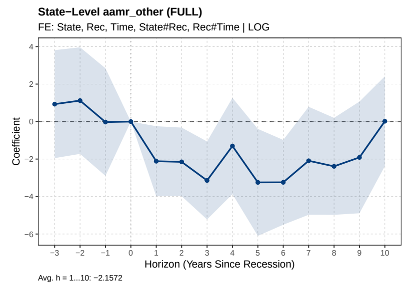
Figure 32: GR Heterogeneity by Race (UR Shock, State Level)



(a) White

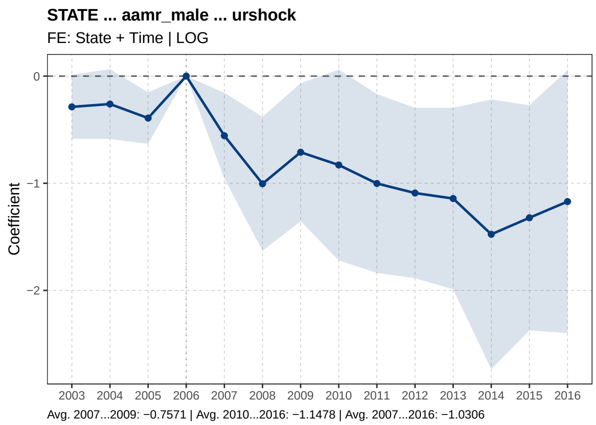


(b) Black

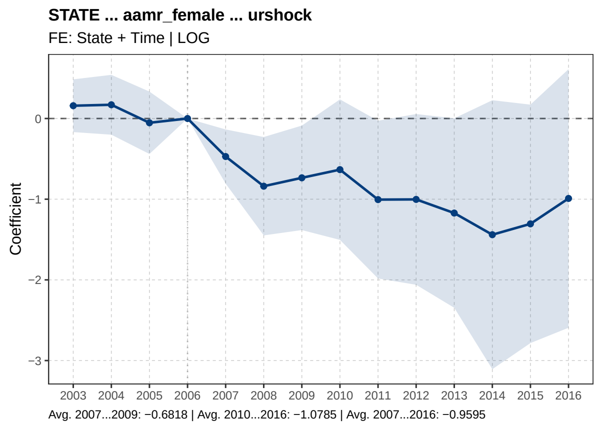


(c) Other

Figure 33: Full Panel Heterogeneity by Race (UR Shock, State Level)

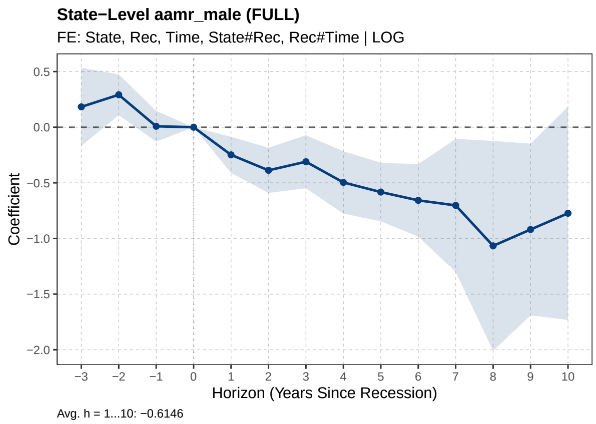


(a) Male

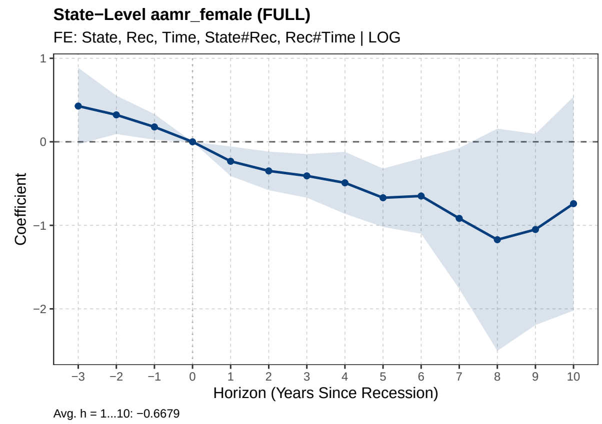


(b) Female

Figure 34: GR Heterogeneity by Sex (UR Shock, State Level)

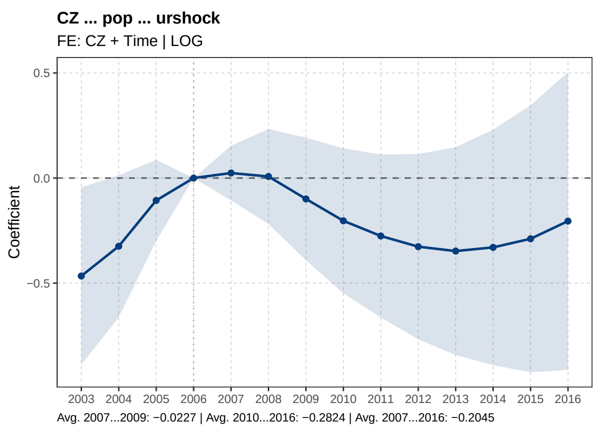


(a) Male

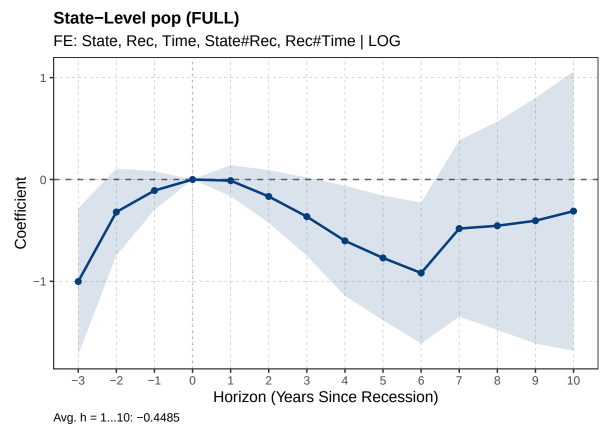


(b) Female

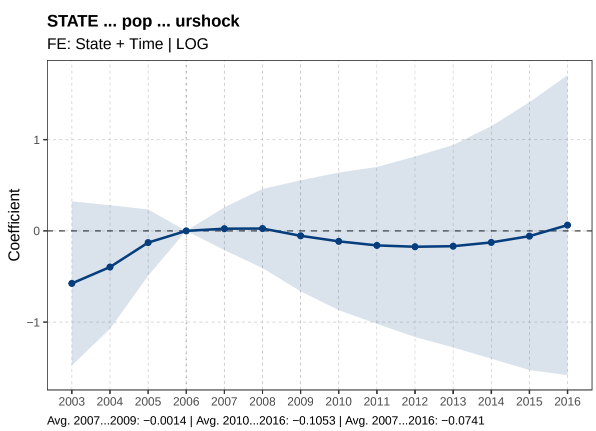
Figure 35: Full Panel Heterogeneity by Sex (UR Shock, State Level)



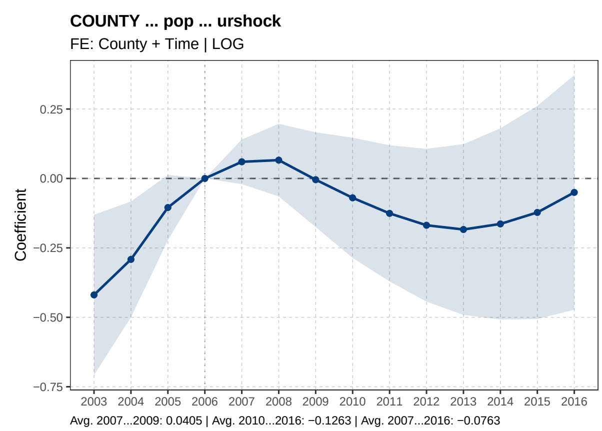
(a) CZ Level, Great Recession



(b) State Level, Full Panel

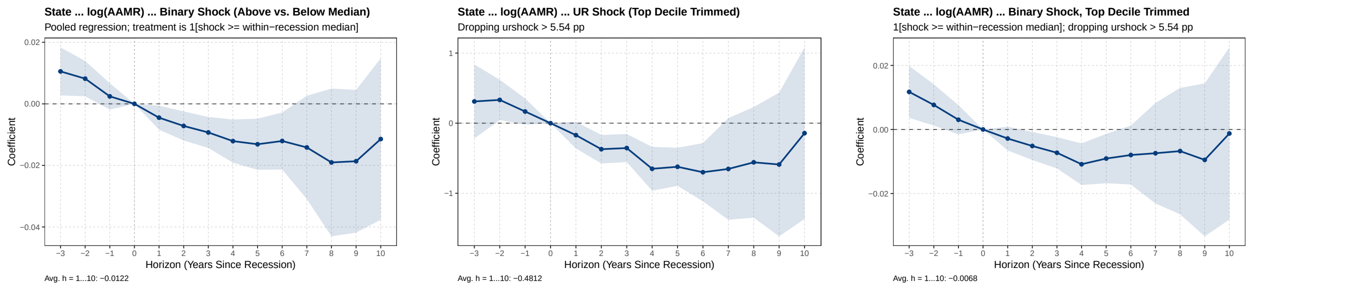


(c) State Level, Great Recession



(d) County Level, Great Recession

Figure 36: Population Event Studies

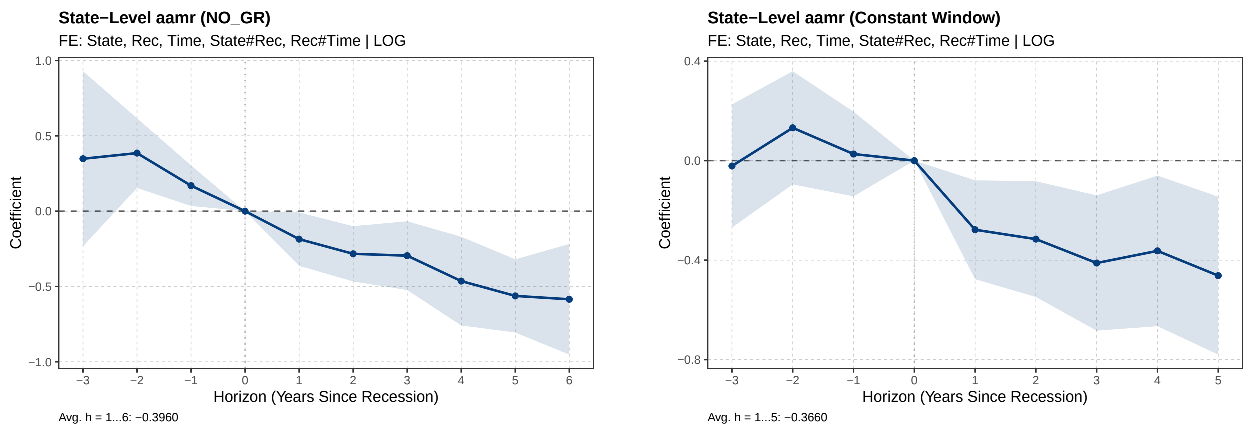


(a) Binary Shock

(b) Trimmed Top Decile

(c) Trimmed Top Decile, Binary Shock

Figure 37: Linearity Checks



(a) Excluding Great Recession

(b) Constant Window

Figure 38: Stacked Event Study Robustness: UR Shock on log(AAMR)

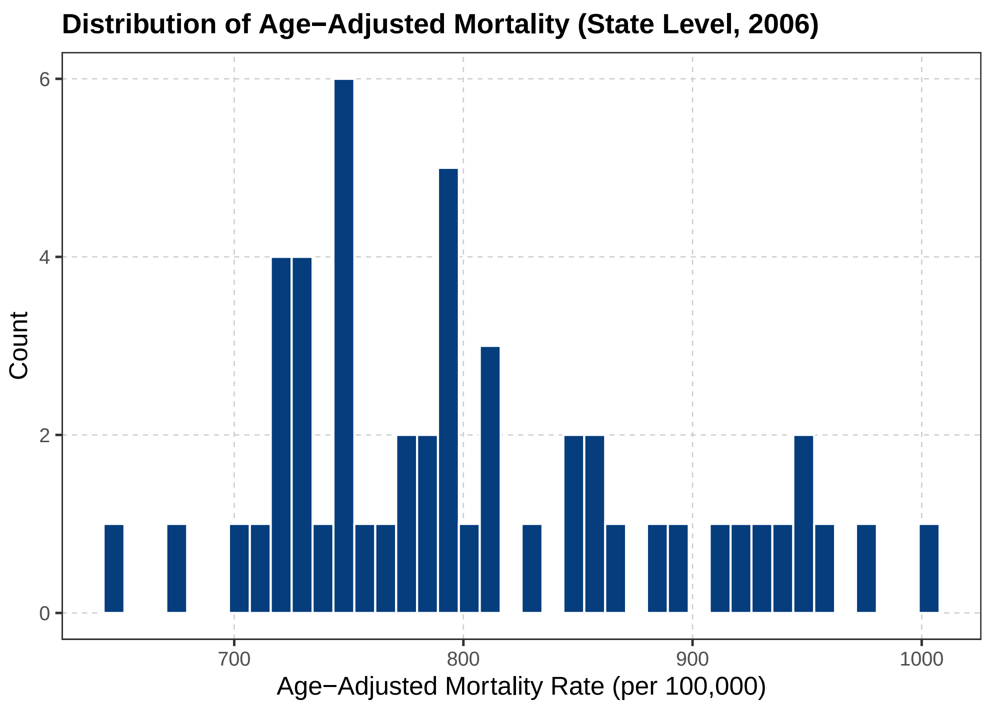


Figure 39: Distribution of AAMR (State Level)

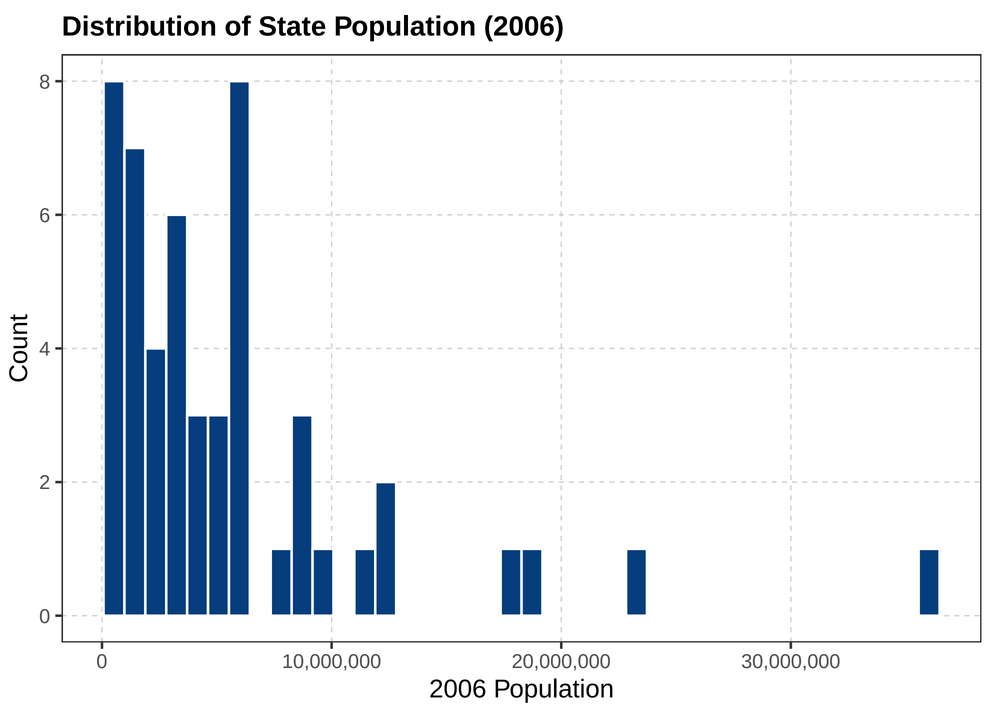


Figure 40: Distribution of State Populations

Table 6: Average Coefficients by Cause of Death (UR Shock)

	Great Recession		6 Recessions
	2007–2009	2010–2016	h = 1–10
External Causes	-0.7950*	-2.4856*	-1.4977**
	(0.4508)	(1.3657)	(0.6821)
Cardiovascular & Metabolic	-0.8919***	-0.9319	-0.4619
	(0.3327)	(0.6492)	(0.2959)
Cancer	-0.0122	-0.1821	-0.0384
	(0.1636)	(0.2319)	(0.1310)
Mental Disorders	-1.5811	-4.7074**	-2.6849**
	(1.0932)	(2.2961)	(1.2254)
Respiratory	-1.3134*	-1.8494	-0.9392
	(0.7866)	(1.5713)	(0.7111)
Infectious & Blood	-0.8349	-1.8747	-1.5570**
	(0.7856)	(1.5474)	(0.6867)
Digestive & Genitourinary	-0.6436	-1.1709	-0.9293***
	(0.4246)	(0.7162)	(0.3507)
Neurological	0.0151	1.3990	0.7500
	(0.5069)	(1.2887)	(0.6591)
Maternal & Perinatal	0.4963	-1.7100	-0.1710
	(0.9514)	(1.6855)	(1.0323)
Other & Ill-Defined	-2.6379	-2.2471	-3.6459**
	(2.2678)	(1.9004)	(1.4307)

Table 7: Average Coefficients by Age Group (UR Shock)

	Great Recession		6 Recessions
	2007–2009	2010–2016	h = 1–10
< 1	0.3846	-0.5146	-1.4059***
	(0.6642)	(0.7723)	(0.3925)
1–4	-4.1243***	-3.9357***	-2.3360***
	(0.8195)	(1.3724)	(0.7631)
5–14	-2.7178	-3.7413*	-0.9648
	(1.7540)	(2.0817)	(0.7508)
15–24	-1.3511	-2.9384*	-1.4869**
	(0.9172)	(1.6147)	(0.6167)
25–34	-0.2611	-1.5480	-1.8657***
	(0.5694)	(1.2939)	(0.7000)
35–44	-1.1654***	-1.9593*	-1.3580**
	(0.3614)	(1.0980)	(0.5624)
45–54	-0.6846*	-2.2624**	-1.1485**
	(0.3777)	(1.0312)	(0.4597)
55–64	-0.2807	-0.3128	-0.3443
	(0.2241)	(0.6522)	(0.3130)
65–74	-0.5949**	-0.7207	-0.4632
	(0.2466)	(0.6936)	(0.2952)
75–84	-0.8809***	-1.3766**	-0.6894***
	(0.2560)	(0.5576)	(0.2571)
85+	-0.7571**	-0.9486	-0.4631*
	(0.3637)	(0.5789)	(0.2755)

Table 8: Recession Event Windows

Cohort	Recession	Reference Year	Non-Overlapping Window	Constant Window
1	1970–1971	1969	1969–1971	1966–1974
2	1974–1975	1973	1972–1976	1970–1978
3	1980–1982	1979	1977–1985	1976–1984
4	1990–1991	1989	1986–1995	1986–1994
5	2001–2002	1999	1996–2002	1996–2004
6	2007–2009	2006	2003–2016	2003–2011

Notes: Reference year is the last pre-recession year ($\tau = 0$). Non-overlapping windows are constructed so that adjacent recession cohorts do not share calendar years; the 1980 and 1981–82 recessions are combined into a single cohort. The constant window uses a fixed $\tau \in \{-3, \ldots, +5\}$ for all cohorts regardless of overlap.